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Primer-BLAST

» **JOB ID:sLptkyNULvwJwivHJqcP9Vy8HsdxrwXacA**

Primer-BLAST Results

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•

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Input PCR template

none

Specificity of primers

Target templates were found in selected database: Core nucleotide BLAST database (Organism limited to Hypocreales)

Other reports

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Detailed primer reports + −

You can re-search for specific primers by accepting some of the unintended targets, check the box(es) next to the ones you accept and try again to re-search for specific primers Submit

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Primer pair 1

	Sequence (5'->3')	Length	Tm	GC%	Self complementarity	Self 3' complementarity
Forward primer	GCATCGATGAAGAACGCAGC	20	60.32	55.00	8.00	3.00
Reverse primer	NTTTCCTCCGCTTATTGATATGC	24	58.77	39.13	6.00	6.00

Products on intended targets

Products on allowed targets

Products on allowed transcript variants

Products on potentially unintended templates

Products on target templates

>[KC461438.1](#) Uncultured Hypocreales clone Casp_0.54_161_13 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template      32 ..... 51

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
Template      390 C..... 367
```

>[AB872440.1](#) Trichoderma sp. 1212-03 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: 1212-03

product length = 374

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      270    ..... 289

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      643    C..... 620

```

>[NR_148078.1](#) *Mariannaea humicola* CBS 740.95 ITS region; from TYPE material

```

product length = 354
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      224    ..... 243

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      577    C..... 554

```

>[AF205452.1](#) *Stachybotrys echinata* strain ATCC22697 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 370
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      353    ..... 372

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      722    C..... 699

```

>[OM106825.1](#) *Fusarium* sp. isolate GLBRC638 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      90     ..... 109

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      430    C..... 407

```

>[ON209615.1](#) *Fusarium poae* strain CN082E1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      310    ..... 329

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      650    C..... 627

```

>[MK280817.1](#) *Fusarium citri* strain LC7922 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      160    ..... 179

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24

```

Template 501 C..... 478

>[MG274317.1](#) *Fusarium venenatum* isolate 11020 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 219 238

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 559 C..... 536

>[OM515026.1](#) *Trichoderma asperelloides* strain CEN1514 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 332 351

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 695 C..... 672

>[JN995626.1](#) *Nectria pseudotrichia* strain GJS091329 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 236 255

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 596 C..... 573

>[JF449532.1](#) Uncultured *Gibberella* clone AC_2w_E06 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 227 246

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 581 C..... 558

>[NR_157491.1](#) *Paracremonium binnewijzendii* CBS 143277 ITS region; from TYPE material

product length = 346

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 303 322

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 648 C..... 625

>[FN859414.1](#) *Cladobotryum protrusum* genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain TFC 201316

product length = 371

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 225

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
Template 576 C..... 553

>[JX244047.1](#) *Neonectria* sp. 216c internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 172 191

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
Template 523 C..... 500

>[HE649414.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT17

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 229

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
Template 572 C..... 549

>[KC305356.1](#) *Stachybotrys chartarum* voucher HGUP 0497 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 307 326

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
Template 672 C..... 649

>[KP899626.1](#) *Neobarya parasitica* voucher Marson s/n 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 370

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 246 265

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
Template 615 C..... 592

>[MW299191.1](#) *Fusarium* sp. isolate 880264-31 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 134 ..... 153

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 488 C..... 465

```

>[OL832175.1](#) *Fusarium acuminatum* strain NRRL 13403 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 176 ..... 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 532 C..... 509

```

>[OL832179.1](#) *Fusarium avenaceum* strain FRC R6679 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 ..... 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515

```

>[HF546325.2](#) *Geosmithia* sp. CCF4223 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF4223, isolate U404

```

product length = 347
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 515 C..... 492

```

>[OQ434117.1](#) *Trichoderma* sp. isolate NB112 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 358
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 183 ..... 202

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

```

>[HF546259.1](#) *Geosmithia pallida* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U112

```

product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 225 ..... 244

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 589 C..... 566

>[MT435072.1](#) *Fusarium algeriense* strain NRRL_66647_F_algeriense internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 536 C..... 513

>[NR_189360.1](#) *Claviceps yanagawaensis* TNS F-96440 ITS region; from TYPE material

product length = 381
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 246 265
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 626 C..... 603

>[OU720036.1](#) *Claviceps truncatispora* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CCC1399

product length = 372
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 206 225
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 577 C..... 554

>[KX958035.1](#) *Clonostachys rosea* isolate 26R-8-F03 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 190 209
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 549 C..... 526

>[MT557121.1](#) *Fusarium* sp. isolate RL96 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 149 168
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 490 C..... 467

>[JN235303.1](#) *Fusarium solani* strain FRC S-2508 5.8S ribosomal RNA gene, complete sequence

product length = 361

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	180		199

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	540	C.....	517

>[OM106443.1](#) *Fusarium* sp. isolate GLBRC245 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	207		226

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	547	C.....	524

>[AY986959.1](#) *Ophiocordyceps sinensis* isolate UASWS0046 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	239		258

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	593	C.....	570

>[MZ423803.1](#) *Hypocreaceae* sp. isolate GW20_637 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	213		232

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	574	C.....	551

>[OL832186.1](#) *Fusarium avenaceum* strain NRRL 36069 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	184		203

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	538	C.....	515

>[KT878335.1](#) *Sarocladium kiliense* strain 02524 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       266  .....                285

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       624  C.....                601

```

>[KM242368.1](#) *Bionectria* sp. INBio_4515M internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       32   .....                51

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       392  C.....                369

```

>[KT878344.1](#) *Acremonium* sp. 06699 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       279  .....                298

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       632  C.....                609

```

>[OM515089.1](#) *Trichoderma rifaii* strain CEN202 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       349  .....                368

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       710  C.....                687

```

>[HG937146.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09J10C92 (SMOTU266)

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       240  .....                259

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       581  C.....                558

```

>[EF026005.1](#) *Lecanicillium lecanii* strain ARSEF No. 6543 18S ribosomal RNA gene, internal transcribed spacer 1, 5.8S ribosomal RNA gene, internal transcribed spacer 2, and 28S ribosomal RNA gene, complete sequence

```

product length = 352
Forward primer  1    GCATCGATGAAGAACGCAGC   20

```

```

Template      199 ..... 218

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      550 C..... 527

```

>[ON411490.1](#) *Fusarium* sp. isolate R6.10 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      140 ..... 159

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      482 C..... 459

```

>[KY484972.2](#) *Fusarium solani* isolate CH-5P internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      186 ..... 205

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      547 C..... 524

```

>[LC535970.1](#) *Trichoderma viride* VMB21 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```

product length = 359
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      604 ..... 623

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      962 C..... 939

```

>[MN153935.1](#) *Fusarium* sp. isolate 120 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      185 ..... 204

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      526 C..... 503

```

>[HE649425.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT53

```

product length = 363
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      210 ..... 229

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24

```

Template 572 C..... 549

>[LC146729.1](#) *Ilyonectria destructans* f. sp. *panacis* genes for 18S ribosomal RNA, internal transcribed spacer 1, 5.8S ribosomal RNA, internal transcribed spacer 2, 28S ribosomal RNA, partial and complete sequence, strain: NBRC 31881

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 178 197

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 531 C..... 508

>[HQ020476.1](#) *Gibberella avenacea* isolate F061 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 171 190

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 525 C..... 502

>[KF918575.1](#) *Fusarium solani* strain DE46 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 207 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 567 C..... 544

>[MH348943.1](#) *Fusarium solani* isolate seqID1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 178 197

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 516

>[KT878353.1](#) *Sarocladium implicatum* strain 04035 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 267 286

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 633 C..... 610

>[KM231763.1](#) *Neonectria tsugae* strain CBS 788.69 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 ..... 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 560 C..... 537
```

>[KU837760.1](#) *Epichloe* sp. strain A251I internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 368
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 207 ..... 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 574 C..... 551
```

>[ON931756.1](#) *Fusarium decemcellulare* strain F10 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 347
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 234 ..... 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 580 C..... 557
```

>[JN235256.1](#) *Fusarium petroliphilum* strain FRC S-2525 5.8S ribosomal RNA gene, complete sequence

```
product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 ..... 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521
```

>[HG936867.1](#) Uncultured *Nectria* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09D10C77 (MOTU90)

```
product length = 360
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 260 ..... 279

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 619 C..... 596
```

>[MK407778.1](#) Uncultured *Fusarium* clone D2008ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 232 ..... 251

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

```

>[MT557518.1](#) *Fusarium oxysporum* isolate RL510 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 140 ..... 159

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 480 C..... 457

```

>[PP725435.1](#) *Hypomyces* sp. voucher GB933 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 237 ..... 256

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 596 C..... 573

```

>[OP237229.1](#) *Fusarium verticillioides* strain 82N small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 551 -..... 529

```

>[OQ513907.1](#) *Clonostachys rosea* strain DTO_461-H3 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 322 ..... 341

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 678 C..... 655

```

>[OM106245.1](#) *Fusarium* sp. isolate GLBRC43 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      185    ..... 204

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      545    C..... 522

```

>[MF070493.1](#) *Corinectria fuckeliana* isolate CPC_25913 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 356
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      234    ..... 253

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      589    C..... 566

```

>[MK280835.1](#) *Fusarium sulawense* strain LC6928 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      160    ..... 179

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      501    C..... 478

```

>[JN235248.1](#) *Fusarium petroliphilum* strain FRC S-2503 5.8S ribosomal RNA gene, complete sequence

```

product length = 364
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      181    ..... 200

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      544    C..... 521

```

>[KM460936.1](#) *Clonostachys rosea* 18S ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      245    ..... 264

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      601    C..... 578

```

>[KF428636.1](#) *Neonectria* sp. nwa_besc_86d internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      172    ..... 191

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      523    C..... 500

```

>[OL439237.1](#) *Thyronectria ulmi* strain TCH1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 361 380

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 724 C..... 701

>[EU307897.1](#) *Metarhizium anisopliae* var. *anisopliae* strain LRC 181 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 559 578

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 925 C..... 902

>[MH582406.1](#) *Fusarium staphyleae* strain NRRL 22316 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 159 178

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 518 C..... 495

>[MK211243.1](#) *Fusarium pseudocircinatum* isolate SF_4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 167 186

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 509 C..... 486

>[KC007311.1](#) *Hypocreales* sp. O_1_LILB_26_5k internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 194 213

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 556 C..... 533

>[PV992418.1](#) *Fusarium oxysporum* isolate >QLHBJ-1-3 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 3 22

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 343 C..... 320

>[EU091007.1](#) *Fusarium oxysporum* isolate F30 internal transcribed space 1 gene, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 176 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 516 C..... 493

>[KF897901.1](#) *Fusarium solani* strain DE35 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 205 224

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 565 C..... 542

>[ON931699.1](#) *Fusarium* sp. strain NRRL 36214 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 347

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 235 254

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 581 C..... 558

>[GU055710.1](#) Uncultured *Nectria* clone NG_R_H12 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 353

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 233 252

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 585 C..... 562

>[MH550497.1](#) *Clonostachys* sp. strain daef27 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 363 382

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 719 C..... 696

>[OM106400.1](#) *Fusarium* sp. isolate GLBRC202 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 213 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 566 C..... 543

>[KF918568.1](#) *Fusarium solani* strain DE26 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 201 220

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 561 C..... 538

>[KR094440.1](#) *Fusarium equiseti* strain G328 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 236 255

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 577 C..... 554

>[ON931740.1](#) *Fusarium* sp. strain NRRL 22506 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 347

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 234 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 580 C..... 557

>[JX243764.1](#) *Neonectria* sp. P. tricho VII internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 166 185

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 519 C..... 496

>[KP263102.1](#) *Nectriaceae* sp. B1A0092P302CC615 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 169 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 530 C..... 507

>[KC007300.1](#) *Nectriaceae* sp. C_2_BESC_319n internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 167 186

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 520 C..... 497

>[KM030314.1](#) *Fusarium oxysporum* strain NRRL 66003 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 153 172

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 493 C..... 470

>[PP336497.1](#) *Trichoderma alni* strain SZMC 28000 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 379 398

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 740 C..... 717

>[MW299224.1](#) *Fusarium* sp. isolate 880334-03 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       138      ..... 157

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       492      C..... 469
```

>[LC793069.1](#) *Lecanicillium* sp. CKSI15-F14-E1-i1 genes for ITS1, 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```
product length = 354
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       248      ..... 267

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       601      C..... 578
```

>[OP556598.1](#) *Fusarium oxysporum* isolate N3 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       179      ..... 198

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       519      C..... 496
```

>[KC691584.1](#) *Fusarium pseudensiforme* strain NRRL 46517 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 360
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       155      ..... 174

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       514      C..... 491
```

>[AF009185.2](#) *Gibberella avenacea* internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence, and 28S ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       171      ..... 190

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       525      C..... 502
```

>[HG936115.1](#) Uncultured *Fusidium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09SWuC08 (MOTU18)

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 249 ..... 268

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 612 C..... 589

```

>[EU307913.1](#) *Metarhizium anisopliae* var. *anisopliae* isolate GBX3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 484 ..... 503

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 850 C..... 827

```

>[OM106264.1](#) *Fusarium* sp. isolate GLBRC63 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 183 ..... 202

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 523 C..... 500

```

>[MT557491.1](#) *Fusarium* sp. isolate RL433 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 141 ..... 160

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 497 C..... 474

```

>[JN943354.1](#) *Hypocrea atroviridis* strain NBRC 8436 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 234 ..... 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 600 C..... 577

```

>[OM106480.1](#) *Fusarium* sp. isolate GLBRC282 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341

```

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 546 C..... 523

>[MK280791.1](#) *Fusarium ipomoeae* strain LC12166 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 161 180

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 502 C..... 479

>[LC831996.1](#) *Fusarium fujikuroi* BYM-4 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 531 C..... 508

>[HE649485.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT96

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 207 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 569 C..... 546

>[EF453193.1](#) *Fusarium sulawesiense* strain NRRL 43730 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 179 198

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 520 C..... 497

>[KT347159.1](#) *Fusarium* sp. isolate FEV51A internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 203 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24

Template 557 C..... 534

>[MT557128.1](#) *Fusarium* sp. isolate RL840 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 145 164

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 499 C..... 476

>[HG935535.1](#) Uncultured *Verticillium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09D50C82 (MOTU6)

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 240 259

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 601 C..... 578

>[MH582450.1](#) *Fusarium* sp. strain NRRL 25084 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 562 C..... 539

>[MZ425961.1](#) *Trichoderma* sp. isolate GW1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 231 250

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 595 C..... 572

>[MH299334.1](#) *Sarocladium zeae* voucher CPO 10.585 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 192 211

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 550 C..... 527

>[JX220370.1](#) *Fusarium* sp. J/Rs/13/3 18S ribosomal RNA, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA, partial sequence

```
product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      152 .....   171

Reverse primer 1   NTTTCCTCCGCTTATTGATATGC   24
Template      492 C.....   469
```

>[MW299199.1](#) *Trichoderma* sp. isolate 880323-10 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 359
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      210 .....   229

Reverse primer 1   NTTTCCTCCGCTTATTGATATGC   24
Template      568 C.....   545
```

>[L07139.1](#) *Epichloe festucae* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 369
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      208 .....   227

Reverse primer 1   NTTTCCTCCGCTTATTGATATGC   24
Template      576 C.....   553
```

>[KC007238.1](#) *Neonectria* sp. C_BESC_184v internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 354
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      169 .....   188

Reverse primer 1   NTTTCCTCCGCTTATTGATATGC   24
Template      522 C.....   499
```

>[MK027217.1](#) *Fusarium* sp. isolate F2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 343
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      211 .....   230

Reverse primer 1   NTTTCCTCCGCTTATTGATATGC   24
Template      552 -.....   530
```

>[MK280807.1](#) *Fusarium luffae* strain LC12167 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 160 179

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 501 C..... 478

>[OM106830.1](#) *Sarocladium strictum* isolate GLBRC643 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 62 81

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 426 C..... 403

>[OQ729801.1](#) *Mariannaea humicola* isolate CNUFC-RD174 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 224 243

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 577 C..... 554

>[KP899627.1](#) *Lichenobarya usneae* voucher Driscoll 1037 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 254 273

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 615 C..... 592

>[OR335875.1](#) *Emericellopsis* sp. strain SLF 0218.0908 p32 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 195 214

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 554 C..... 531

>[MW365369.1](#) *Fusarium irregulare* strain VTCC:31704 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 175 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 516 C..... 493

>[KY607535.1](#) *Ijuhya vitellina* strain 41ED internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2', partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 365 384

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 727 C..... 704

>[MW299184.1](#) *Fusarium* sp. isolate 880264-19 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 173 192

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 527 C..... 504

>[KC305355.1](#) *Stachybotrys chartarum* voucher HGUP 0810 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 371

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 311 330

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 681 C..... 658

>[KJ542170.1](#) *Metarhizium brunneum* isolate Billerica internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 147 166

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 515 C..... 492

>[LT598648.1](#) *Lecanicillium* sp. GTG12-11 genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, isolate GTG12-11

```

product length = 354
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       217  .....                236

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       570  C.....                547

```

>[HG936118.1](#) Uncultured *Fusidium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09SWuC32 (MOTU18)

```

product length = 364
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       249  .....                268

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       612  C.....                589

```

>[HQ115647.1](#) *Myrothecium roridum* strain CBS 331.51 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 379
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       233  .....                252

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       611  C.....                588

```

>[MH550473.1](#) *Fusarium* sp. strain daef3 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       360  .....                379

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       720  C.....                697

```

>[MZ742821.1](#) *Fusarium sulawesiense* isolate DG1205193 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       79   .....                98

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template       420  C.....                397

```

>[HF546385.1](#) *Fusarium brachygibbosum* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain TUEF26

```

product length = 351
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template       155  .....                174

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 505 C..... 482

>[JX435189.1](#) *Fusarium cf. solani* CBS 2012 strain CBS 124892 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 181 200
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 542 C..... 519

>[MH107922.1](#) *Varicosporellopsis aquatilis* culture CBS:143509 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 348
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 570 C..... 547

>[GU170644.1](#) *Fusarium keratoplasticum* strain NRRL 46438 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 192 211
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 553 C..... 530

>[OQ799962.1](#) *Fusarium oxysporum* f. sp. *ciceris* isolate MPFOC36 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 164 183
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 504 C..... 481

>[KM242312.1](#) *Hypomyces* sp. INBio_4505I internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 32 51
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24

Template 397 C..... 374

>[JN235541.1](#) *Fusarium cf. dimerum* a DPGS-2011 strain FRC E-329 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 108 127

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 466 C..... 443

>[GU183129.1](#) *Myrothecium verrucaria* strain NRRL 52420 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521

>[KR094454.1](#) *Trichoderma brevicompactum* strain G374 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 237 256

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 593 C..... 570

>[KM231754.1](#) *Mariannaea pinicola* strain CBS 745.88 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 223 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 C..... 553

>[KT876565.1](#) *Fusarium dimerum* isolate R5-2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 170 189

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 528 C..... 505

>[MT557199.1](#) *Fusarium* sp. isolate RL834 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 150 169

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 510 C..... 487

>[MT889756.1](#) *Simplicillium* sp. isolate 002-D8 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 370

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 339 358

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 708 C..... 685

>[DQ459820.1](#) *Gibberella zeae* strain NRRL28336 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 562 C..... 539

>[KF897896.1](#) *Fusarium solani* strain BRID43 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 209 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 569 C..... 546

>[PV590690.1](#) *Fusarium oxysporum* isolate GLBRC1323 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 216 235

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 556 C..... 533

>[MF817199.1](#) *Fusarium equiseti* voucher UNI PIENSA J internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 187 206

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 528 C..... 505

>[LC158825.1](#) *Trichoderma harzianum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: TM2-1

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 233 252

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 594 C..... 571

>[HF546222.1](#) *Geosmithia putterillii* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U131a

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 222 241

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 582 C..... 559

>[KM231784.1](#) *Sarcopodium flavolanatum* strain CBS 128370 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 239 258

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 599 C..... 576

>[LC598885.1](#) *Claviceps humidiphila* TNS:F-60498 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 238 257

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 606 C..... 583

>[OR467037.1](#) *Trichoderma reesei* strain LE13 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 358
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       368 ..... 387

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       725 C..... 702

```

>[MH055398.1](#) *Fusarium oxysporum* isolate DSM 106834 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       247 ..... 266

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       587 C..... 564

```

>[MN543999.1](#) *Trichoderma* sp. isolate CLE146 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       236 ..... 255

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       597 C..... 574

```

>[EU871030.1](#) *Hypocrea lixii* strain TU Graz 10TSMC1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       242 ..... 261

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       603 T..... 580

```

>[AJ301997.1](#) *Myrothecium cinctum* 18S rRNA gene, 5.8S rRNA gene, 28S rRNA gene (partial), internal transcribed spacer 1 (ITS1) and internal transcribed spacer 2 (ITS2), strain BBA 69182

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       1972 ..... 1991

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       2323 C..... 2300

```

>[KJ570711.1](#) *Thyronectria roseovirens* strain MA4 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC   20

```

```

Template      336 ..... 355

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      687 C..... 664

```

>[KT876566.1](#) *Fusarium dimerum* isolate R8-2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```

product length = 359
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      172 ..... 191

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      530 C..... 507

```

>[MW199163.1](#) *Fusarium keratoplasticum* strain BD1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      187 ..... 206

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      548 C..... 525

```

>[JF740913.1](#) *Fusarium* sp. NRRL 52717 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      136 ..... 155

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      477 C..... 454

```

>[MW298949.1](#) *Hypocreales* sp. isolate 836478-26--4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      184 ..... 203

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      544 C..... 521

```

>[JF740926.1](#) *Fusarium torulosum* strain NRRL 52772 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      169 ..... 188

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24

```

Template 525 C..... 502

>[OR019722.1](#) *Fusarium* sp. isolate F36 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 216 235

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 557 C..... 534

>[MT892802.1](#) *Emericellopsis maritima* isolate 001-B5 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 308 327

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 668 C..... 645

>[JN235324.1](#) *Fusarium* cf. *solani* 18-c DPGS-2011 strain FRC S-2438 5.8S ribosomal RNA gene, complete sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 180 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

>[MH685567.1](#) *Stachybotrys* sp. strain CNUFC-GW2S-4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 189 208

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 555 C..... 532

>[GU982311.1](#) *Gibberella moniliformis* strain PUMCH10XB00173 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 209 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 550 -..... 528

>[KC461334.1](#) Uncultured Hypocreales clone Ntet_1.5_37_48 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 341

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template       32 ..... 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template       372 C..... 349
```

>[NR_137650.1](#) Clonostachys grammicospora CBS 209.93 ITS region; from TYPE material

product length = 358

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template       190 ..... 209

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template       547 C..... 524
```

>[PV590735.1](#) Sarocladium strictum isolate GLBRC1381 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template       104 ..... 123

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template       468 C..... 445
```

>[OQ513885.1](#) Purpureocillium lilacinum strain DTO_460-I8 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 353

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template       364 ..... 383

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template       716 C..... 693
```

>[OM515061.1](#) Trichoderma hamatum strain CEN1350 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template       332 ..... 351

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template       693 C..... 670
```

>[HE604164.1](#) Geosmithia sp. MK1772 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain MK1772

```

product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 ..... 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 C..... 553

```

>[EU307910.1](#) *Metarhizium anisopliae* var. *anisopliae* isolate D1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 368
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 500 ..... 519

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 867 C..... 844

```

>[OM515020.1](#) *Trichoderma afroharzianum* strain CEN289 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 350 ..... 369

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 711 C..... 688

```

>[KX681609.1](#) *Fusarium solani* isolate H06A_1F_1a internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 124 ..... 143

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 484 C..... 461

```

>[KF918572.1](#) *Fusarium solani* strain DE37 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 208 ..... 227

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 568 C..... 545

```

>[MN153963.1](#) *Fusarium* sp. isolate 44 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      185    ..... 204

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      526    C..... 503

```

>[KR906687.1](#) *Fusarium solani* isolate U1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      37      ..... 56

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      398    C..... 375

```

>[KJ572165.1](#) *Epichloe* sp. MJ-2014 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 370
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      207    ..... 226

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      576    C..... 553

```

>[MT557264.1](#) *Fusarium* sp. isolate RL552 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      143    ..... 162

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      484    C..... 461

```

>[PV935560.1](#) *Trichoderma* sp. strain KMM 4738 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 365
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      362    ..... 381

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      726    C..... 703

```

>[ON207822.1](#) *Acremonium* sp. strain CN053B3 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      324    ..... 343

```

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 682 C..... 659

>[MH481700.1](#) *Parengyodontium album* strain F4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 228 247

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 589 C..... 566

>[ON411474.1](#) *Fusarium oxysporum* isolate F63 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 164 183

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 504 C..... 481

>[KX681575.1](#) *Fusarium fujikuroi* isolate H06A_4K_1b internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 149 168

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 503 C..... 480

>[PX350289.1](#) *Fusarium* cf. *solani* isolate LLC1812 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 220 239

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 581 C..... 558

>[MZ573413.1](#) *Fusarium* sp. voucher G.M. 2014-12-07.5 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 328 347

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24

Template 683 C..... 660

>[JN235532.1](#) *Fusarium* cf. *dimerum* a DPGS-2011 strain FRC E-319 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 107 126

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 465 C..... 442

>[JF740897.1](#) *Fusarium* sp. NRRL 25134 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 523 C..... 500

>[OM515041.1](#) *Trichoderma azevedoi* strain CEN1242 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 350 369

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 712 C..... 689

>[MN450211.1](#) *Gliocephalotrichum brasiliense* strain CCUB381 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 310 329

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 664 C..... 641

>[FJ545305.1](#) *Metarhizium anisopliae* isolate CNYN2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 548 567

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 916 C..... 893

>[OM515032.1](#) *Trichoderma asperelloides* strain CEN277 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 332 351

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 695 C..... 672

>[KM012000.1](#) *Cosmospora* sp. NRRL 62961 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 533 C..... 510

>[OR558367.1](#) *Catenomargarita pseudocercosporicola* strain COAD 3662 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 204 223

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 569 C..... 546

>[OL832320.1](#) *Fusarium avenaceum* strain NRRL 54939 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 538 C..... 515

>[PX350070.1](#) *Fusarium* sp. isolate LLC1503 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 219 238

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 569 C..... 546

>[MT557235.1](#) *Trichoderma* sp. isolate RL74 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 185 204

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 549 C..... 526

>[JF740883.1](#) *Fusarium citri* strain NRRL 25084 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 176 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 517 C..... 494

>[KT878327.1](#) *Acremonium sclerotigenum* strain 01844 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 259 278

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 620 C..... 597

>[JX435213.1](#) *Fusarium* cf. *solani* CBS 2012 strain CBS 115658 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 181 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 543 C..... 520

>[LC681719.1](#) *Claviceps miscanthicola* TNS:F-96450 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 380

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 215 234

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 594 C..... 571

>[HE649476.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT66

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 572 C..... 549

```

>[KC808234.1](#) *Fusarium* sp. FSSC_12g culture NRRL:54968 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 366
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 173 ..... 192

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515

```

>[MT557387.1](#) *Clonostachys* sp. isolate RL437 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 137 ..... 156

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 493 C..... 470

```

>[KX681599.1](#) *Fusarium* cf. *incarnatum* MLST 8-a isolate H05A_1H_4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 124 ..... 143

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 465 C..... 442

```

>[OL617412.1](#) *Calcarisporium arbuscula* strain 11JW2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 238 ..... 257

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 596 -..... 574

```

>[MT557167.1](#) *Fusarium* sp. isolate RL824 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       148    ..... 167

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       489    C..... 466

```

>[OR825838.1](#) *Fusarium equiseti* isolate BAK2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       212    ..... 231

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       553    G..... 530

```

>[ON931745.1](#) *Fusarium* sp. strain NRRL 53937 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 347
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       234    ..... 253

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       580    C..... 557

```

>[MT321486.1](#) *Trichoderma yunnanense* isolate Ty_6_ Maharashtra internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 364
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       216    ..... 235

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       578    -..... 556

```

>[OR536572.1](#) *Fusarium oxysporum* strain VT74 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       183    ..... 202

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       523    C..... 500

```

>[JX313065.1](#) *Beauveria bassiana* isolate TPT-RARS-Bb4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       157    ..... 176

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 508 C..... 485

>[OL719280.1](#) *Fusarium* sp. isolate 1901NT-2.2.2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 286 305
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 645 C..... 622

>[HG996181.1](#) uncultured *Hypocreales* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, clone c3cdfcd1

product length = 347
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 183 202
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 529 C..... 506

>[PX350296.1](#) *Fusarium* sp. isolate LLC1825 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 570 C..... 547

>[PX350203.1](#) *Fusarium* sp. isolate LLC1702 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 231 250
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 584 C..... 561

>[LC792969.1](#) *Lecanicillium* sp. CKSI15-F06-TR-i4 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 354
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 563 C..... 540

>[AB587022.1](#) *Fusarium langsethiae* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: FRC T-0992

```
product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 509 C..... 486
```

>[PX350288.1](#) *Fusarium* cf. *solani* isolate LLC1811 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 ..... 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 582 C..... 559
```

>[XR_007157351.1](#) *Purpureocillium takamizusanense* 28S ribosomal RNA (JDV02_010901), rRNA

```
product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 29 ..... 48

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 380 C..... 357
```

>[MZ423601.1](#) *Nectriaceae* sp. isolate GW20_119 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 520 C..... 497
```

>[GU797142.1](#) *Fusarium* sp. SS-R3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 338 ..... 357

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 698 C..... 675
```

>[AB114631.1](#) *Ephelis japonica* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain:NIAES 6584

```
product length = 387
```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      1958  ..... 1977

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      2344  C..... 2321

```

>[OU720011.1](#) *Claviceps lovelessii* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CCC1413

```

product length = 370
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      221  ..... 240

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      590  C..... 567

```

>[OM515078.1](#) *Trichoderma longibrachiatum* strain CEN1281 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      373  ..... 392

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      732  C..... 709

```

>[PX350196.1](#) *Fusarium* sp. isolate LLC1695 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 351
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      220  ..... 239

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      570  C..... 547

```

>[MH299340.1](#) *Sarocladium zeae* voucher CPO 10.635 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      172  ..... 191

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      530  C..... 507

```

>[HM055441.1](#) *Metarhizium robertsii* strain ARSEF 967 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      305  ..... 324

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 671 C..... 648

>[MT316380.1](#) *Trichoderma longibrachiatum* strain DTO 403-D5 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 398 417

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 756 C..... 733

>[MT557221.1](#) *Trichoderma gamsii* isolate RL316 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 172 191

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 536 C..... 513

>[AB517937.1](#) *Acremonium* sp. CH019 gene for 28S ribosomal RNA, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 34 53

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 393 C..... 370

>[MW791945.1](#) *Trichoderma paraviridescens* voucher culture Y114I internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 587 C..... 564

>[GQ505439.1](#) *Fusarium* sp. NRRL 34012 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 579 C..... 556

>NR_163316.1 *Parasarocladium debuynii* CBS 144942 ITS region; from TYPE material

product length = 362

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	226		245

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	587	C.....	564

>KC461350.1 Uncultured *Hypocreales* clone Casp_0.54_161_14 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 347

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	32		51

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	378	C.....	355

>LT746271.1 *Fusarium solani* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CPC 27198

product length = 361

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	223		242

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	583	C.....	560

>HG996231.1 uncultured *Hypocreales* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, clone d232e134

product length = 351

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	193		212

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	543	C.....	520

>EU082104.1 *Fusarium* sp. M6 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	167		186

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	508	C.....	485

>LC333211.1 *Fusicolla acetilerea* RF3 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 356

Forward primer	1	GCATCGATGAAGAACGCAGC	20
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```

Template      163 ..... 182

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC 24
Template      518 C..... 495

```

>[OM106669.1](#) *Fusarium* sp. isolate GLBRC473 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC 20
Template      183 ..... 202

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC 24
Template      523 C..... 500

```

>[KM577645.1](#) *Fusarium begoniae* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1   GCATCGATGAAGAACGCAGC 20
Template      315 ..... 334

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC 24
Template      657 C..... 634

```

>[KT719192.1](#) *Lecanicillium* sp. 57 XC-2016 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC 20
Template      187 ..... 206

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC 24
Template      538 C..... 515

```

>[HG936870.1](#) Uncultured *Nectria* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09S50C48 (MOTU90)

```

product length = 360
Forward primer 1   GCATCGATGAAGAACGCAGC 20
Template      260 ..... 279

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC 24
Template      619 C..... 596

```

>[KF918580.1](#) *Fusarium solani* strain DI12 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1   GCATCGATGAAGAACGCAGC 20
Template      204 ..... 223

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC 24

```

Template 545 C..... 522

>[AB586998.1](#) *Fusarium merismoides* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: MAFF 236504

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 171 190

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 522 C..... 499

>[DQ459871.1](#) *Fusarium pseudograminearum* strain NRRL28062 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 339

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 559 C..... 536

>[PP812519.1](#) *Fusarium vanettenii* strain 77-13-4 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 325 344

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 685 C..... 662

>[HE649456.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT20

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 209 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 570 C..... 547

>[MN644610.1](#) *Trichoderma afroharzianum* voucher research collection Farrer lab 206 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 548 C..... 525

>[PP959271.1](#) *Fusarium* sp. strain UROL F58 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 127 146

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 483 C..... 460

>[PX350265.1](#) *Fusarium* sp. isolate LLC1778 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 218 237

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 571 C..... 548

>[KM242320.1](#) *Hypocrea* sp. INBio_4506K internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 32 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 388 C..... 365

>[JQ411383.1](#) *Clonostachys rosea* isolate B1a027A7ASNA2CC940 internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 175 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 531 C..... 508

>[MZ423759.1](#) *Nectriaceae* sp. isolate GW20_521 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 545 C..... 522

>[OU720007.1](#) *Claviceps* sp. KP-2018c genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CCC1248

product length = 371

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 C..... 553

>[AJ875331.1](#) *Neonectria radicola* 5.8S rRNA gene, ITS1 and ITS2, isolate cy22

product length = 353

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 205 224

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 557 C..... 534

>[NR_157430.1](#) *Cylindrium syzygii* ITS region; from TYPE material

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 219 238

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 561 C..... 538

>[KM231788.1](#) *Ophionectria trichospora* strain CBS 314.75 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 278 297

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 637 C..... 614

>[OM515023.1](#) *Trichoderma asperelloides* strain CEN1338 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 332 351

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 695 C..... 672

>[MK405948.1](#) Uncultured *Fusarium* clone D142ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20

```

Template      250 ..... 269

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      604 C..... 581

```

>[KJ570697.1](#) *Thyronectria lamyi* strain NL5 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      356 ..... 375

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      710 C..... 687

```

>[PX350086.1](#) *Fusarium* sp. isolate LLC1532 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 351
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      219 ..... 238

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      569 C..... 546

```

>[JN235180.1](#) *Fusarium keratoplasticum* strain FRC S-2430 5.8S ribosomal RNA gene, complete sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      54 ..... 73

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      415 C..... 392

```

>[ON411501.1](#) *Fusarium* sp. isolate T1.15 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      179 ..... 198

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      521 C..... 498

```

>[MT557203.1](#) *Fusarium* sp. isolate RL625 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      147 ..... 166

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24

```

Template 488 C..... 465

>[HQ591385.1](#) *Cordyceps militaris* strain DYU1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 350

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 408 427

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 757 C..... 734

>[JN235200.1](#) *Fusarium keratoplasticum* strain FRC S-2466 5.8S ribosomal RNA gene, complete sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 543 C..... 520

>[JX083518.1](#) *Claviceps purpurea* isolate CCC597 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 580 C..... 557

>[OL832188.1](#) *Fusarium avenaceum* strain NRRL 53729 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515

>[LR586046.1](#) *Neonectria* sp. PJ-2019 genomic DNA sequence contains 18S rRNA gene, ITS1, isolate *Neonectria* sp. pj30 ex *Placidium* sp. PJ-2019a

product length = 346

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 41 60

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 386 C..... 363

>[PP947326.1](#) *Clonostachys cf. eriocamporesii* strain CN071D7 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 358 ..... 377

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 714 C..... 691
```

>[PP345625.1](#) *Fusarium commune* isolate TT internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 180 ..... 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 534 C..... 511
```

>[OM106512.1](#) *Fusarium sp.* isolate GLBRC314 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 201 ..... 220

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 541 C..... 518
```

>[JN235317.1](#) *Fusarium petroliphilum* strain FRC S-2512 5.8S ribosomal RNA gene, complete sequence

```
product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 ..... 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521
```

>[EU091019.1](#) *Fusarium oxysporum* isolate F5 internal transcribed space 1 gene, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 176 ..... 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 517 C..... 494
```

>[MT557239.1](#) *Clonostachys sp.* isolate RL431 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit

ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 156 175

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 512 C..... 489

>[KM066570.2](#) *Trichoderma* sp. BAB-3580 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 263 282

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 625 C..... 602

>[PP336533.1](#) *Fusarium lateritium* strain NRRL 13622 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 326 345

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 677 C..... 654

>[KF367481.1](#) *Trichoderma* sp. 3 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 309 328

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 670 C..... 647

>[EF027376.1](#) Uncultured *Hypocreales* clone bg1a internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 353

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 529 C..... 506

>[OQ589890.1](#) *Trichoderma* sp. strain 008_A7 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 362 ..... 381

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 724 C..... 701

```

>[LC158828.1](#) *Trichoderma asperellum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: V76-14

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 215 ..... 234

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 578 C..... 555

```

>[FJ545322.1](#) *Metarhizium anisopliae* isolate NLHN2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 369
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 548 ..... 567

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 916 C..... 893

```

>[OM106463.1](#) *Fusarium* sp. isolate GLBRC265 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 205 ..... 224

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 545 C..... 522

```

>[MT453299.1](#) *Stachybotryaceae* sp. isolate DSM101519_C11_RLCS23 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 379
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 262 ..... 281

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 640 C..... 617

```

>[AB378508.1](#) *Lecanicillium attenuatum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: KYK00039

```

product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 230 ..... 249

```

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 581 C..... 558

>[HG936664.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10JWuC83 (MOTU53)

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 246 265

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 587 C..... 564

>[OL832288.1](#) *Fusarium reticulatum* strain FRC T683 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 538 C..... 515

>[MK280837.1](#) *Fusarium guilinese* strain LC12160 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 160 179

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 501 C..... 478

>[JN943361.1](#) *Hypocrea lutea* strain NBRC 9061 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 251 270

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 607 C..... 584

>[MH361214.1](#) *Fusarium culmorum* isolate Kh4 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 345

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 215 234

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 559 C..... 536

>[HM534900.1](#) *Thyronectria sinopica* strain CBS 127386 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       333  .....  352

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       684  C.....  661
```

>[JN943316.1](#) *Elaphocordyceps ophioglossoides* strain NBRC 8992 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```
product length = 351
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       210  .....  229

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       560  C.....  537
```

>[AB360362.1](#) *Lecanicillium saksenae* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: BTCC-F20

```
product length = 354
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       1982  .....  2001

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       2335  C.....  2312
```

>[MT557481.1](#) *Clonostachys* sp. isolate RL513 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       140  .....  159

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       496  C.....  473
```

>[MN095347.1](#) *Fusarium oxysporum* isolate F21 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       163  .....  182

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       503  C.....  480
```

>[HE974453.1](#) *Fusarium oxysporum* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain SK 1649

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 164 ..... 183

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 504 C..... 481

```

>[JQ649345.1](#) *Hypocreales* sp. B1A0812PD2CC482 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 108 ..... 127

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 474 C..... 451

```

>[FJ545297.1](#) *Metarhizium anisopliae* isolate CNHN3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 368
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 547 ..... 566

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 914 C..... 891

```

>[OL832277.1](#) *Fusarium reticulatum* strain FRC R8712 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 ..... 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515

```

>[HM484540.1](#) *Nectria balsamea* strain A.R. 4478 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 353
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 189 ..... 208

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 541 C..... 518

```

>[JX282603.1](#) *Fusarium solani* isolate BR11 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 364

```

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 ..... 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521
```

>[MN310200.1](#) Gliomastix tumulicola isolate P6462 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 369
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 211 ..... 230

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 579 C..... 556
```

>[PP937578.1](#) Fusarium sp. strain S1.1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 360
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 299 ..... 318

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 658 C..... 635
```

>[KM066579.2](#) Trichoderma sp. BAB-3730 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 366
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 219 ..... 238

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 584 C..... 561
```

>[PV593214.1](#) Trichoderma sp. isolate GLBRC1446 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 227 ..... 246

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 591 C..... 568
```

>[FN859397.1](#) Cladobotryum cubitense genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain G.A. m643.w

```
product length = 368
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 208 ..... 227
```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 575 C..... 552

>[KC341964.1](#) *Fusarium* sp. AL-19 IRH-2012f internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 144 163
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 485 C..... 462

>[JN235157.1](#) *Fusarium keratoplasticum* strain FRC S-2411 5.8S ribosomal RNA gene, complete sequence

product length = 362
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 543 C..... 520

>[KX359614.1](#) *Fusarium* cf. *solani* isolate SAN1077_2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 577 C..... 554

>[MN644690.1](#) *Fusarium equiseti* voucher research collection Farrer lab 16 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 183 202
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 524 C..... 501

>[PQ343891.1](#) *Emericellopsis* sp. isolate UROLF254 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 577 C..... 554

>[MT557137.1](#) *Fusarium* sp. isolate RL659 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       146      ..... 165

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       486      C..... 463
```

>[AM421042.1](#) *Geosmithia putterillii* 18S rRNA gene (partial), 5.8S rRNA gene, 28S rRNA gene (partial), ITS1 and ITS2, strain CCF3342

```
product length = 361
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       191      ..... 210

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       551      C..... 528
```

>[HG940472.1](#) *Hypocrea nigricans* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain FALH26, isolate FALH26

```
product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       273      ..... 292

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       634      C..... 611
```

>[HF546250.1](#) *Geosmithia langdonii* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U91

```
product length = 365
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       220      ..... 239

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       584      C..... 561
```

>[MK280792.1](#) *Fusarium arcuatisporum* strain LC6026 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       161      ..... 180

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       502      C..... 479
```

>[ON411510.1](#) *Fusarium proliferatum* isolate T5.3 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      123 .....   142

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      477 C.....   454

```

>[EU750689.1](#) *Fusarium* sp. 19003 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      322 .....   341

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      683 C.....   660

```

>[NR_173406.1](#) *Fusicolla aquaeductuum* ITS region; from TYPE material

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      225 .....   244

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      576 C.....   553

```

>[JN387049.1](#) *Trichoderma atroviride* strain TUCIM N154 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      215 .....   234

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      581 T.....   558

```

>[MH990628.1](#) *Simplicillium sympodiophorum* isolate C3A 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 374
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      391 .....   410

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      764 C.....   741

```

>[MK622797.1](#) *Trichoderma* sp. isolate PMK internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      254 .....   273

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 610 C..... 587

>[LC793007.1](#) *Lecanicillium* sp. CKSI15-F10-E1-i1 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 564 C..... 541

>[OM515022.1](#) *Trichoderma asperelloides* strain CEN1277 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 332 351

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 695 C..... 672

>[OM106605.1](#) *Fusarium* sp. isolate GLBRC408 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 183 202

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 537 C..... 514

>[OM106694.1](#) *Fusarium* sp. isolate GLBRC498 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 209 228

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 549 C..... 526

>[PV590670.1](#) *Trichoderma koningiopsis* isolate GLBRC1302 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 244 263

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24

Template 608 C..... 585

>[ON931719.1](#) *Fusarium* sp. strain KOD 1237 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 347

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 234 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 580 C..... 557

>[L07136.1](#) *Epichloe glyceriae* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 205 224

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 572 C..... 549

>[KC806268.1](#) *Clonostachys byssicola* strain CML 1943 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 174 193

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 530 C..... 507

>[OM502981.1](#) *Fusarium solani* isolate KJ1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 171 190

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 532 C..... 509

>[MK408154.1](#) Uncultured *Fusarium* clone D2385ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 250 269

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 604 C..... 581

>[FN859420.1](#) *Cladobotryum tenue* genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CBS 152.92

product length = 370

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 203 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 572 C..... 549

>[OM735575.1](#) *Sarocladium zeae* voucher IL00111_C370-7 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 95 114

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 454 C..... 431

>[KM030307.1](#) *Nectria haematococca* strain NRRL 62977 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 136 155

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 496 C..... 473

>[EF453078.1](#) *Fusarium* sp. NRRL 43435 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 543 C..... 520

>[OM515050.1](#) *Trichoderma brevicompactum* strain CEN1245 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 351 370

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 707 C..... 684

>[MH858143.1](#) *Emericellopsis pusilla* culture CBS:226.62 strain CBS 226.62 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene,

and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 334 353

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 694 C..... 671

>[MT557247.1](#) *Fusarium* sp. isolate RL598 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 147 166

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 488 C..... 465

>[PP947371.1](#) *Fusarium* sp. 2 CD-2024 strain CN106G5 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 337 356

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 678 C..... 655

>[HF546264.1](#) *Geosmithia flava* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U124

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

>[FJ545286.1](#) *Metarhizium anisopliae* isolate CNGD11 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 546 565

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 913 C..... 890

>[KM231813.1](#) *Fusarium sambucinum* strain CBS 146.95 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 339
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      221 .....   240

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      559 C.....   536

```

>[HG995536.1](#) uncultured Hypocreales genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, clone 1164018c

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      199 .....   218

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      550 C.....   527

```

>[KX664333.1](#) Trichoderma reesei isolate F07-01 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 358
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      278 .....   297

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      635 C.....   612

```

>[KU846755.1](#) Striatibotrys eucylindrosporus strain CBS 203.61 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 371
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      239 .....   258

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      609 C.....   586

```

>[NR_185397.1](#) Claviceps arundinellae CCC 1240 ITS region; from TYPE material

```

product length = 371
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      206 .....   225

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      576 C.....   553

```

>[KY327805.1](#) Metarhizium flavoviride isolate A1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 397
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      201 .....   220

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24

```

Template 597 T..... 574

>[LT746269.1](#) *Fusarium solani* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CPC 27192

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 224 243

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 584 C..... 561

>[HG996224.1](#) uncultured *Hypocreales* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, clone d1905048

product length = 353

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 203 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 555 C..... 532

>[FJ765013.1](#) *Isaria fumosorosea* isolate NLHG-2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 256 275

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 607 C..... 584

>[MW298969.1](#) *Fusarium* sp. isolate 836490-09 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 137 156

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 492 C..... 469

>[JF449884.1](#) Uncultured *Bionectria* clone SW_2w_G03 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 232 251

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 588 C..... 565

>[AB378512.1](#) *Lecanicillium attenuatum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: NBRC 103238

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 225 244

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 C..... 553

>[JN387048.1](#) *Hypocrea lixii* strain TUCIM N156 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 T..... 521

>[OM106469.1](#) *Fusarium* sp. isolate GLBRC271 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 207 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 547 C..... 524

>[KT719187.1](#) *Lecanicillium* sp. 4 XC-2016 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 187 206

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515

>[FJ545298.1](#) *Metarhizium anisopliae* isolate CNHN4 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 547 566

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 914 C..... 891

>[OP810851.1](#) *Fusarium solani* isolate FNVS01 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2,

complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        229  .....                248

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        589  C.....                566
```

>[EU091013.1](#) *Fusarium oxysporum* isolate F41 internal transcribed space 1 gene, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        165  .....                184

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        505  C.....                482
```

>[HF546354.2](#) *Geosmithia langdonii* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U300

```
product length = 365
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        220  .....                239

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        584  C.....                561
```

>[MZ423717.1](#) *Nectriaceae* sp. isolate GW20_423 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        179  .....                198

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        519  C.....                496
```

>[KM066588.2](#) *Fusarium* sp. BAB-3607 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        209  .....                228

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        550  C.....                527
```

>[MT557122.1](#) *Fusarium* sp. isolate RL68 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        143  .....          162

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        484  C.....          461

```

>[KY484984.2](#) *Fusarium solani* isolate FPOST-164 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        185  .....          204

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        545  C.....          522

```

>[MZ425501.1](#) *Clonostachys rosea* isolate DG-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        148  .....          167

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        504  C.....          481

```

>[OR019737.1](#) *Fusarium* sp. isolate F20 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        210  .....          229

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        551  C.....          528

```

>[ON411472.1](#) *Fusarium oxysporum* isolate F52 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        180  .....          199

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        520  C.....          497

```

>[KT878342.1](#) *Acremonium sclerotigenum* strain 06457 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      259      ..... 278

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      620      C..... 597

```

>[MW007885.1](#) *Trichoderma asperellum* isolate TriRice small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 364
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      244      ..... 263

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      606      -..... 584

```

>[ON931754.1](#) *Fusarium decemcellulare* strain F02 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 347
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      232      ..... 251

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      578      C..... 555

```

>[MH858186.1](#) *Emericellopsis microspora* culture CBS:380.62 strain CBS 380.62 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      334      ..... 353

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      693      C..... 670

```

>[HG936683.1](#) Uncultured *Neonectria* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09J10C69 (MOTU56)

```

product length = 354
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      220      ..... 239

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      573      C..... 550

```

>[ON411475.1](#) *Fusarium solani* isolate F64 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      177      ..... 196

```

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 538 C..... 515

>[HE649406.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT4

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 213 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 575 C..... 552

>[LC545403.1](#) *Simplicillium sympodiophorum* FCG1835 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 374

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 239 258

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 612 C..... 589

>[KU974299.1](#) *Fusarium* sp. FSSC isolate F53 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 167 186

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 531 C..... 508

>[KF367524.1](#) *Trichoderma* sp. 10 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 335 354

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 697 C..... 674

>[AY833600.1](#) *Torrubiella confragosa* voucher UASWS0012 internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 211 230

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 562 C..... 539

>[HG936547.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10J10C17 (MOTU42)

```
product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 252 ..... 271

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 603 C..... 580
```

>[ON193777.1](#) *Fusarium acuminatum* strain UNIJAG.PL.OP1188 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 204 ..... 223

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 560 C..... 537
```

>[OQ422046.1](#) *Fusarium* sp. breed MHUB11465 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 203 ..... 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 564 T..... 541
```

>[MW301195.1](#) *Myrothecium inundatum* isolate F11(NSA3&NLC2) small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 378
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 316 ..... 335

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 693 C..... 670
```

>[NR_145022.1](#) *Hypomyces robledoi* TFC 2007-17 ITS region; from TYPE material

```
product length = 373
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 225 ..... 244

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 597 C..... 574
```

>[JN943369.1](#) *Hypocrea nigricans* strain NBRC 30611 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 247 ..... 266

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 609 C..... 586

```

>[OL830714.1](#) *Fusarium oxysporum* isolate YL001 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 175 ..... 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 515 T..... 492

```

>[OM106249.1](#) *Fusarium* sp. isolate GLBRC47 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 185 ..... 204

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 545 C..... 522

```

>[MT557257.1](#) *Fusarium* sp. isolate RL777 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 146 ..... 165

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 486 C..... 463

```

>[PP959316.1](#) *Trichoderma* sp. isolate UROL_F151 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 129 ..... 148

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 491 C..... 468

```

>[OM106556.1](#) *Fusarium* sp. isolate GLBRC359 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341

```

```

Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      165  ..... 184

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC  24
Template      505  C..... 482

```

>[KT876549.1](#) *Fusarium solani* isolate R1-3 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```

product length = 364
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      168  ..... 187

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC  24
Template      531  C..... 508

```

>[MN954349.1](#) *Fusarium rekanum* isolate CMW54723 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      198  ..... 217

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC  24
Template      557  C..... 534

```

>[JX010680.1](#) *Fusarium* sp. AD-2012a strain cbe1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      151  ..... 170

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC  24
Template      491  -..... 469

```

>[LC793061.1](#) *Lecanicillium* sp. CKSI15-F13-TL-i1 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```

product length = 355
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      209  ..... 228

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC  24
Template      563  C..... 540

```

>[KC007159.1](#) *Neonectria* sp. BESC196a internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 353
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      170  ..... 189

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC  24

```

Template 522 C..... 499

>**PV593218.1** *Trichoderma* sp. isolate GLBRC1451 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 235 254

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 599 C..... 576

>**HF546277.1** *Geosmithia* sp. U154 genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U154

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 180 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 543 C..... 520

>**HG935540.1** Uncultured *Verticillium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09J70C60 (MOTU6)

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 240 259

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 601 C..... 578

>**PV485098.1** *Metarhizium anisopliae* isolate IIHRMa13 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 372

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 205 224

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 T..... 553

>**EU307890.1** *Metarhizium anisopliae* var. *anisopliae* strain LRC 112 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 325 344

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 691 C..... 668

>[MK408069.1](#) Uncultured *Fusarium* clone D2300ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 232 251

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 585 C..... 562

>[KF367557.1](#) *Trichoderma* sp. 16 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 314 333

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 678 C..... 655

>[GQ167226.1](#) *Sarocladium zeae* culture NRRL:34559 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 198 217

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 556 C..... 533

>[GQ167230.1](#) *Fusarium proliferatum* culture-collection NRRL:6413 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 193 212

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 535 C..... 512

>[OR752062.1](#) *Fusarium iranicum* isolate KJ96 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 224 243

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 578 C..... 555

>[KC007242.1](#) Nectriaceae sp. C_BESC_184z internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 167 186

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 521 C..... 498

>[ON931743.1](#) Fusarium sp. strain NRRL 22513 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 347

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 234 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 580 C..... 557

>[KF918576.1](#) Fusarium solani strain DEB6 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 207 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 569 C..... 546

>[OM106519.1](#) Fusarium sp. isolate GLBRC321 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 199 218

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 539 C..... 516

>[AM421039.1](#) Geosmithia fassatiae 18S rRNA gene (partial), 5.8S rRNA gene, 28S rRNA gene (partial), ITS1 and ITS2, strain AK31/98

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 213 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 578 C..... 555

>[MW298955.1](#) Fusarium sp. isolate 836489-10 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit

ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 136 155

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 490 C..... 467

>[LC793054.1](#) *Lecanicillium* sp. CKSI15-F13-E1-i2 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 564 C..... 541

>[KY413717.1](#) *Trichoderma* sp. isolate ERS035 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 358

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 237 256

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 594 C..... 571

>[MW298970.1](#) *Fusarium* sp. isolate 836490-12 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 188 207

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 542 C..... 519

>[KC311482.1](#) *Fusarium culmorum* isolate E13b 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 219 238

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 560 C..... 537

>[MZ423665.1](#) *Nectriaceae* sp. isolate GW20_298 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 520 C..... 497

```

>[KU974300.1](#) *Fusarium* sp. FSSC isolate 1554 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 165 ..... 184

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 526 C..... 503

```

>[LC793055.1](#) *Lecanicillium* sp. CKSI15-F13-E1-i3 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 564 C..... 541

```

>[EU082096.1](#) *Fusarium* sp. M14 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 163 ..... 182

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 523 C..... 500

```

>[AB378519.1](#) *Lecanicillium psalliotae* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: KYK00031

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 239 ..... 258

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 592 C..... 569

```

>[PP336538.1](#) *Fusarium concolor* strain NRRL 13459 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 323 ..... 342

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 665 C..... 642

>[KF367471.1](#) *Purpureocillium* sp. 1 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 364 383

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 715 C..... 692

>[KM066571.2](#) *Trichoderma* sp. BAB-3614 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 238 257

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 596 C..... 573

>[KF918590.1](#) *Fusarium solani* strain DIR63 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 205 224

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 565 C..... 542

>[JX243967.1](#) *Neonectria* sp. 33i internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 353

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 172 191

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 524 C..... 501

>[MZ423722.1](#) *Nectriaceae* sp. isolate GW20_431 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24

Template 542 C..... 519

>[KM066576.2](#) *Trichoderma* sp. BAB-3544 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 253 272

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 615 C..... 592

>[GU170654.1](#) *Fusarium* sp. NRRL 53132 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 155 174

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 516 C..... 493

>[ON399101.1](#) *Trichoderma virens* strain T2C1.4 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 358

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 1969 1988

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 2326 C..... 2303

>[JN943367.1](#) *Hypocrea muroiana* strain NBRC 31288 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 225 244

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 591 C..... 568

>[KT876644.1](#) *Fusarium oxysporum* isolate C3-4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 166 185

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 506 C..... 483

>[PP336505.1](#) *Trichoderma* sp. SZMC 28011 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 376 395

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 739 C..... 716

>[ON411495.1](#) *Fusarium nygamai* isolate T1.3 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 158 177

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 511 C..... 488

>[ON193838.1](#) *Trichoderma atroviride* strain UNIJAG.PL.OP643 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 596 C..... 573

>[ON209611.1](#) *Fusarium poae* strain CN082D2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 343 362

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 683 C..... 660

>[LT548277.1](#) *Lecanicillium* sp. CCF 5233 genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF 5233

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 1924 1943

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 2285 C..... 2262

>[KJ499909.1](#) *Clonostachys pseudochroleuca* strain CML 2513 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and

28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 175 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 531 C..... 508

>[EU860061.1](#) *Cosmospora aurantiicola* strain F-251,305 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 565 C..... 542

>[KM225680.1](#) *Thyronectria caudata* strain NCA1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 333 352

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 687 C..... 664

>[JQ922211.1](#) *Nectriaceae* sp. B1a040-2-SNA2 isolate PanB1a0402SNA2CC25 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 540 C..... 517

>[LR586017.1](#) *Neonectria* sp. PJ-2019 genomic DNA sequence contains 18S rRNA gene, ITS1, isolate *Neonectria* sp. pj28 ex *Placidium* sp. PJ-2019a

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 41 60

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 392 C..... 369

>[GQ505685.1](#) *Fusarium nanum* strain NRRL 22244 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 222 ..... 241

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 563 C..... 540

```

>[KF367487.1](#) *Trichoderma* sp. 7 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 334 ..... 353

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 696 C..... 673

```

>[OQ421960.1](#) *Fusarium* sp. breed MHUB11355 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 199 ..... 218

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 559 T..... 536

```

>[KU504283.1](#) *Trichoderma* sp. BAB-3297 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 252 ..... 271

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 614 C..... 591

```

>[FJ589649.1](#) *Metarhizium anisopliae* isolate SPCT1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 548 ..... 567

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 914 C..... 891

```

>[PX060179.1](#) *Trichoderma viride* isolate ARE_AAFC_7.A.2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      345      ..... 364

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      704      C..... 681

```

>[AB378557.1](#) *Hirsutella proturicola* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```

product length = 389
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      265      ..... 284

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      653      C..... 630

```

>[OU720026.1](#) *Claviceps* sp. KP-2018d genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CCC1458

```

product length = 375
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      168      ..... 187

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      542      C..... 519

```

>[HG798769.1](#) *Trichoderma harzianum* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain TUH125

```

product length = 363
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      111      ..... 130

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      473      C..... 450

```

>[HE649447.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT87

```

product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      206      ..... 225

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      567      C..... 544

```

>[KC808242.1](#) *Fusarium falciforme* culture NRRL:54977 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      178      ..... 197

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      540      C..... 517

```

>[KF897897.1](#) *Fusarium solani* strain DE2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 208 227

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 571 C..... 548

>[OM106514.1](#) *Fusarium* sp. isolate GLBRC316 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 546 C..... 523

>[AF081472.2](#) *Stachybotrys dichroa* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 548 567

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 914 C..... 891

>[MW757244.1](#) *Mariannaea fusiformis* isolate CNUFC YJ7 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 126 145

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 477 C..... 454

>[KC461354.1](#) Uncultured Hypocreales clone Absp_4.3_118_35 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 32 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 393 C..... 370

>[PQ570877.1](#) *Trichoderma merleae* strain MST FP3586 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 166 185

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 530 C..... 507

>[GQ505738.1](#) *Fusarium longifundum* strain NRRL 36372 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 222 241

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 563 C..... 540

>[GU017492.1](#) *Hypocreales* sp. KH00223 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 242 261

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 605 C..... 582

>[MT612345.1](#) *Fusarium graminearum* isolate Fg-W11 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 344

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 213 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 556 T..... 533

>[AY188916.1](#) *Gibberella fujikuroi* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 208 227

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 550 C..... 527

>[MF276664.1](#) *Trichoderma longibrachiatum* isolate 0501G small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 299 318

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 658 C..... 635

>[KM056334.1](#) *Paecilomyces marquandii* strain NRRL 62995 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 380

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 149 168

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 528 C..... 505

>[KM242365.1](#) *Myrothecium* sp. INBio_4515l internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 32 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 390 C..... 367

>[LC793009.1](#) *Lecanicillium* sp. CKSI15-F10-E1-i3 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 564 C..... 541

>[MH857151.1](#) *Emericellopsis mirabilis* culture CBS:177.53 strain CBS 177.53 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 334 353

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 693 C..... 670

>[PX350287.1](#) *Fusarium cf. solani* isolate LLC1810 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 221 240

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
 Template 582 C..... 559

>[MW298894.1](#) *Hypocreales* sp. isolate 836445-02 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 138 157

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
 Template 498 C..... 475

>[MT078886.1](#) *Keithomyces* sp. DT-2020a strain CBS 158.69 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 382

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 376 395

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
 Template 757 C..... 734

>[OM106745.1](#) *Fusarium* sp. isolate GLBRC552 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 137 156

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
 Template 477 C..... 454

>[PV590679.1](#) *Trichoderma* sp. isolate GLBRC1312 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 257 276

Reverse primer 1 NTTTTCCTCCGCTTATTGATATGC 24
 Template 621 C..... 598

>[PX091242.1](#) *Fusarium nygamai* isolate 0724_570_033_PCR_RX5_ITS_1_H03.ab1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 354
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       183      ..... 202

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       536      T..... 513
```

>[ON209609.1](#) *Fusarium poae* strain CN082C9 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       361      ..... 380

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       701      C..... 678
```

>[OM106670.1](#) *Fusarium* sp. isolate GLBRC474 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       183      ..... 202

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       523      C..... 500
```

>[OQ589874.1](#) *Fusarium acuminatum* strain 006_E4 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       319      ..... 338

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       675      C..... 652
```

>[HG935523.1](#) Uncultured *Verticillium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09D10C25 (MOTU6)

```
product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       240      ..... 259

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       601      C..... 578
```

>[MT557427.1](#) *Fusarium* sp. isolate RL829 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit

ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 142 161

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 483 C..... 460

>[XR_001936385.1](#) *Fusarium oxysporum* f. sp. *lycopersici* 4287 28S ribosomal RNA (FOXG_19923), rRNA

product length = 340

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 29 48

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 368 C..... 345

>[LC792913.1](#) *Lecanicillium* sp. CKSI15-F02-TR-i3 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 152 171

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 506 C..... 483

>[MT557157.1](#) *Fusarium* sp. isolate RL750 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 146 165

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 502 C..... 479

>[NR_167685.1](#) *Epichloe hybrida* ATCC TSD-66 ITS region; from TYPE material

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 298 317

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 665 C..... 642

>[NR_185401.1](#) *Sphaerostilbella toxica* TFC 202258 ITS region; from TYPE material

product length = 375

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24

Template 597 C..... 574

>[NR_148077.1](#) *Mariannaea pinicola* CBS 745.88 ITS region; from TYPE material

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 223 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 C..... 553

>[JQ411369.1](#) *Trichoderma reesei* isolate M20312EM2CC1162 internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 358

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 254 273

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 611 C..... 588

>[KF367513.1](#) *Fusarium* sp. 1 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 323 342

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 663 C..... 640

>[OM515037.1](#) *Trichoderma atroviride* strain CEN875 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 334 353

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 699 C..... 676

>[NR_155358.1](#) *Memnoniella longistipitata* ATCC 22699 ITS region; from TYPE material

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 288 307

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 655 C..... 632

>[ON074989.1](#) *Fusarium chlamydosporum* strain CN026B1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1    GCATCGATGAAGAACGCAGC    20
Template       372 .....    391

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC    24
Template       728 C.....    705

```

>[OM515007.1](#) *Trichoderma afroharzianum* strain CEN1249 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1    GCATCGATGAAGAACGCAGC    20
Template       351 .....    370

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC    24
Template       712 C.....    689

```

>[MW299261.1](#) *Fusarium* sp. isolate 880345-30-11 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1    GCATCGATGAAGAACGCAGC    20
Template       138 .....    157

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC    24
Template       480 C.....    457

```

>[MW299189.1](#) *Fusarium* sp. isolate 880264-26 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1    GCATCGATGAAGAACGCAGC    20
Template       132 .....    151

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC    24
Template       486 C.....    463

```

>[OM515016.1](#) *Trichoderma afroharzianum* strain CEN235 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1    GCATCGATGAAGAACGCAGC    20
Template       350 .....    369

Reverse primer 1    NTTTTCCTCCGCTTATTGATATGC    24
Template       711 C.....    688

```

>[OQ422014.1](#) *Fusarium* sp. breed MHUB11433 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342

```

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 203 ..... 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 T..... 521
```

>[MH299329.1](#) *Sarocladium zeae* voucher CPO 10.527 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 173 ..... 192

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 531 C..... 508
```

>[OL832218.1](#) *Fusarium* sp. FTSC_10 strain FRC R7722 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 179 ..... 198

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 535 C..... 512
```

>[LT746255.1](#) *Fusarium sarcochrom* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CPC 26369

```
product length = 351
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 224 ..... 243

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 574 C..... 551
```

>[KC461381.1](#) Uncultured *Hypocreales* clone Dasp_1.3_22_1 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 32 ..... 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 393 C..... 370
```

>[LT746244.1](#) *Fusarium* sp. 2 MSD-2017 genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CPC 27709

```
product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 224 ..... 243

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
```

Template 580 C..... 557

>[FJ765026.1](#) *Paecilomyces marquandii* isolate SKCH-5 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 380

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 591 C..... 568

>[PV876367.1](#) *Fusarium delphinoides* isolate GLF2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 214 233

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 569 C..... 546

>[HG940471.1](#) *Hypocrea nigricans* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain FALH12, isolate FALH12

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 274 293

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 635 C..... 612

>[KF880411.1](#) *Fusarium solani* strain DEB28 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 209 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 572 C..... 549

>[OQ803298.1](#) *Fusarium oxysporum* f. sp. *ciceris* strain UPFOC42 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 29 48

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 369 C..... 346

>[HQ718417.1](#) *Gibberella moniliformis* strain Gm3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 343
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      167 ..... 186

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      508 ..... 486
```

>[OM876881.1](#) *Fusarium tricinctum* strain BMT6 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      211 ..... 230

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      567 C..... 544
```

>[KC007214.1](#) *Neonectria* sp. C_BESC_184ab internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 354
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      169 ..... 188

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      522 C..... 499
```

>[HF546220.1](#) *Geosmithia* sp. U307a genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U307a

```
product length = 365
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      219 ..... 238

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      583 C..... 560
```

>[OL872344.1](#) *Hypomyces microspERMUS* strain ATHUM 6866 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 365
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      394 ..... 413

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      758 C..... 735
```

>[MK408149.1](#) Uncultured *Fusarium* clone D2380ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer

2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        233 .....                252

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        586 C.....                563

```

>[OM106661.1](#) *Fusarium* sp. isolate GLBRC465 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        183 .....                202

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        523 C.....                500

```

>[AB823651.1](#) *Myrothecium roridum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: KKFC390

```

product length = 376
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        206 .....                225

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        581 C.....                558

```

>[MW299460.1](#) *Hypocrea* sp. isolate 928717-34 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        135 .....                154

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        495 C.....                472

```

>[PX350304.1](#) *Fusarium* sp. isolate LLC1838 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        218 .....                237

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        559 C.....                536

```

>[ON209707.1](#) *Fusarium* sp. strain CN122C3 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        321  .....          340

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        662  C.....          639

```

>[MT557088.1](#) *Trichoderma* sp. isolate RL551 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        197  .....          216

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        559  C.....          536

```

>[OQ248231.1](#) *Fusarium equiseti* strain CoLD Pav_40 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        337  .....          356

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        678  C.....          655

```

>[PX350050.1](#) *Fusarium* cf. *solani* isolate LLC1476 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        182  .....          201

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        541  C.....          518

```

>[HG965025.1](#) *Sarocladium ochraceum* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene and ITS2, culture collection CBS 428.67

```

product length = 375
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        194  .....          213

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        568  C.....          545

```

>[KC920582.1](#) *Isaria farinosa* isolate YHIFXS-I001 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer  1    GCATCGATGAAGAACGCAGC   20

```

```

Template      2745  ..... 2764

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC 24
Template      3096  C..... 3073

```

>[LC793025.1](#) *Lecanicillium* sp. CKSI15-F11-E1-i3 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```

product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC 20
Template      209    ..... 228

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC 24
Template      563    C..... 540

```

>[KM231854.1](#) *Cylindrium aeruginosum* strain CBS 693.83 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 340
Forward primer 1      GCATCGATGAAGAACGCAGC 20
Template      228    ..... 247

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC 24
Template      567    C..... 544

```

>[FJ545294.1](#) *Metarhizium anisopliae* isolate CNHE 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 369
Forward primer 1      GCATCGATGAAGAACGCAGC 20
Template      547    ..... 566

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC 24
Template      915    C..... 892

```

>[MT135009.1](#) *Fusarium verticillioides* isolate N24R41 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1      GCATCGATGAAGAACGCAGC 20
Template      171    ..... 190

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC 24
Template      513    A..... 490

```

>[PV716402.1](#) *Beauveria bassiana* isolate TP13 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC 20
Template      190    ..... 209

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC 24

```

Template 541 C..... 518

>[MF076600.1](#) *Fusarium* sp. isolate S106 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 371 390

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 712 C..... 689

>[PQ422815.1](#) *Metarhizium* sp. isolate Mero28A01 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 534 -..... 512

>[LT598644.1](#) *Acremonium* sp. TK1-22 genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, isolate TK1-22

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 203 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 561 C..... 538

>[AB378528.1](#) *Lecanicillium* sp. KYK00214 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 238 257

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 588 C..... 565

>[MW659182.1](#) *Fusarium incarnatum* strain F15 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 180 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 520 -..... 498

>[MH685570.1](#) *Memnoniella* sp. strain CNUFC-MSW24-2-12 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 371

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 167 186

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 537 C..... 514

>[OP174987.1](#) *Trichoderma brevicompactum* strain 2IG2102 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 300 319

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 656 C..... 633

>[KJ542161.1](#) *Metarhizium robertsii* isolate M330-08 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 142 161

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 508 C..... 485

>[JF740923.1](#) *Fusarium lacertarum* strain NRRL 52753 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 183 202

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 524 C..... 501

>[MN954335.1](#) *Fusarium mekan* isolate CMW53696 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 197 216

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 556 C..... 533

>[AY857231.1](#) *Trichoderma harzianum* strain TUB F-839 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```
product length = 364
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       226  ..... 245

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       589  T..... 566
```

>[OL832234.1](#) *Fusarium acuminatum* strain FRC R8280 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       182  ..... 201

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       538  C..... 515
```

>[HG937033.1](#) Uncultured *Verticillium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09J70C20 (MOTU156)

```
product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       204  ..... 223

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       560  C..... 537
```

>[MT251174.1](#) *Fusarium falciforme* strain DTO 421-G2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       326  ..... 345

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       687  C..... 664
```

>[HG936687.1](#) Uncultured *Neonectria* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10S70C68 (MOTU56)

```
product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       232  ..... 251

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       586  C..... 563
```

>[MW299459.1](#) *Hypocreales* sp. isolate 928717-32 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       203 .....                222

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       563 C.....                540

```

>[MT557506.1](#) *Fusarium* sp. isolate RL296 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       144 .....                163

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       484 C.....                461

```

>[JN235255.1](#) *Fusarium petroliphilum* strain FRC S-2524 5.8S ribosomal RNA gene, complete sequence

```

product length = 364
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       162 .....                181

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       525 C.....                502

```

>[LT548278.1](#) *Lecanicillium* sp. CCF 5201 genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF 5201

```

product length = 354
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       1919 .....                1938

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       2272 C.....                2249

```

>[KF918583.1](#) *Fusarium solani* strain DI25 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       206 .....                225

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       566 C.....                543

```

>[HG936132.1](#) Uncultured *Fusidium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10JWuC38 (MOTU18)

```

product length = 364
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       249 .....                268

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24

```

Template 612 C..... 589

>[MT892810.1](#) *Trichoderma citrinoviride* isolate 001-D3 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 358

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 356 375

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 713 C..... 690

>[PP897823.1](#) *Fusarium proliferatum* isolate ITS41 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 175 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 529 C..... 506

>[KM030294.1](#) *Acremonium persicinum* strain NRRL 62964 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 547 C..... 524

>[KY413716.1](#) *Calonectria* sp. isolate ERS034 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 227 246

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 580 C..... 557

>[LN846839.1](#) *Claviceps pazoutovae* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CS 352A

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 238 257

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 606 C..... 583

>[OP596045.1](#) *Sarocladium zeae* strain CN097C5 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 359
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       333 ..... 352

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       691 C..... 668
```

>[OM515057.1](#) *Trichoderma erinaceum* strain CEN1558 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 365
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       334 ..... 353

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       698 C..... 675
```

>[FJ545316.1](#) *Metarhizium anisopliae* isolate LSTC2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 368
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       548 ..... 567

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       915 C..... 892
```

>[MW299479.1](#) *Fusarium* sp. isolate 928739-26--11 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       83 ..... 102

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       439 C..... 416
```

>[AY899259.1](#) *Cordyceps militaris* strain CICC14013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 25S ribosomal RNA gene, partial sequence

```
product length = 351
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       188 ..... 207

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       538 C..... 515
```

>[JN995622.1](#) *Nectria balansae* strain AR4635 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 193 212

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 561 C..... 538

>[MK828121.1](#) *Fusarium proliferatum* strain 144 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 944 963

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 1304 C..... 1281

>[HE649405.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT3

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 204 223

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 566 C..... 543

>[HG935553.1](#) Uncultured *Verticillium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10D50C39 (MOTU6)

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 240 259

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 601 C..... 578

>[HF546267.1](#) *Geosmithia* sp. CCF3644 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF3644, isolate U133b

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

>[ON207932.1](#) *Fusarium temperatum* strain CN062F6 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 317 ..... 336

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 659 C..... 636
```

>[MG274306.1](#) *Fusarium equiseti* isolate 10015 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 224 ..... 243

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 565 C..... 542
```

>[AF205451.1](#) *Stachybotrys echinata* strain ATCC34173 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 371
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 353 ..... 372

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 723 C..... 700
```

>[EF453090.1](#) *Fusarium* sp. NRRL 43454 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 178 ..... 197

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 518 C..... 495
```

>[KC461463.1](#) Uncultured *Hypocreales* clone Rhir_4.4_120_12 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 32 ..... 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 390 C..... 367
```

>[PX350256.1](#) *Fusarium* sp. isolate LLC1764 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 218 ..... 237
```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 571 C..... 548

>[OL410542.1](#) *Fusarium solani* isolate Wf12 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 279 298
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 639 C..... 616

>[LC792966.1](#) *Lecanicillium* sp. CKSI15-F06-TR-i1 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 355
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 564 C..... 541

>[KC461313.1](#) Uncultured *Hypocreales* clone Jmex_4.3_115_10 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 341
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 32 51
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 372 C..... 349

>[HG936942.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09J70C73 (MOTU113)

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 246 265
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 587 C..... 564

>[FJ939394.2](#) *Stilbella fimetaria* isolate SES201 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 360
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 576 C..... 553

>[MK407961.1](#) Uncultured *Fusarium* clone D2192ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 232 251

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 585 C..... 562

>[LC663161.1](#) *Fusarium acuminatum* P23-R1-1 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, and 28S rRNA, partial and complete sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 185 204

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 541 C..... 518

>[AJ557830.1](#) *Nectria mauritiicola* 18S rRNA gene (partial), ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene (partial), strain NRRL20420

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 262 281

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 620 C..... 597

>[MN310246.1](#) *Trichoderma* sp. isolate P6541 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 171 190

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 534 C..... 511

>[LT216451.1](#) *Claviceps* sp. CCC 1240 genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCC 1240

product length = 371

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 576 C..... 553

>[KM242291.1](#) *Isaria* sp. INBio_4065D internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 380
Forward primer  1    GCATCGATGAAGAACGCAGC    20
Template        32    .....    51

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC    24
Template        411  C.....    388

```

>[OQ509679.1](#) *Fusarium pseudensiforme* strain 2314 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC    20
Template        226  .....    245

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC    24
Template        586  C.....    563

```

>[GQ505459.1](#) *Fusarium* sp. NRRL 43631 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer  1    GCATCGATGAAGAACGCAGC    20
Template        224  .....    243

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC    24
Template        580  C.....    557

```

>[OQ872065.1](#) *Cordyceps tenuipes* isolate RLC_1687_iNat_131739346 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 351
Forward primer  1    GCATCGATGAAGAACGCAGC    20
Template        251  .....    270

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC    24
Template        601  C.....    578

```

>[FJ545278.1](#) *Metarhizium anisopliae* isolate CNGD3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer  1    GCATCGATGAAGAACGCAGC    20
Template        548  .....    567

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC    24
Template        914  C.....    891

```

>[ON411468.1](#) *Fusarium oxysporum* isolate F24 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      198  .....  217

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      538  C.....  515

```

>[HQ130667.1](#) *Fusarium* sp. WF108 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      209  .....  228

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      570  C.....  547

```

>[HM055439.1](#) *Metarhizium brunneum* strain ARSEF 3297 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 368
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      306  .....  325

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      673  C.....  650

```

>[PV992431.1](#) *Fusarium oxysporum* isolate ZFB-1-2 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      5      .....  24

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      345  C.....  322

```

>[MT557112.1](#) *Clonostachys* sp. isolate RL658 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      156  .....  175

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template      512  C.....  489

```

>[KU872826.1](#) *Fusarium oxysporum* isolate AFIC21 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA region gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      323  .....  342

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 663 C..... 640

>[KX681572.1](#) *Fusarium proliferatum* isolate M03S_A2_91 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 149 168
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 502 C..... 479

>[KP263090.1](#) *Hypocreaceae* sp. M20317P30CC1145 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 360
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 183 202
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 542 C..... 519

>[MZ425964.1](#) *Fusarium fujikuroi* isolate GW4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 132 151
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 474 C..... 451

>[MT557204.1](#) *Fusarium* sp. isolate RL612 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 144 163
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 486 C..... 463

>[HF546285.1](#) *Geosmithia* sp. U154 genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U154, isolate U185B

product length = 364
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 181 200
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24

Template 544 C..... 521

>[PP336499.1](#) *Trichoderma cerinum* strain SZMC 28002 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 376 395

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 736 C..... 713

>[OM106765.1](#) *Trichoderma* sp. isolate GLBRC576 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 102 121

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 466 C..... 443

>[KY024400.1](#) *Fusarium brachygibbosum* strain CA-016 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 159 178

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 508 -..... 486

>[KC461492.1](#) Uncultured *Hypocreales* clone Swis_1.3_24_5 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 339

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 32 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 370 C..... 347

>[LN626654.1](#) *Claviceps purpurea* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, isolate CCC1191

product length = 370

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 581 C..... 558

>[KJ541484.1](#) *Fusarium brachygibbosum* isolate Utah 1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 351
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      224 .....   243

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      574 C.....   551
```

>[OM106277.1](#) *Fusarium* sp. isolate GLBRC77 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      165 .....   184

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      505 C.....   482
```

>[KC461426.1](#) Uncultured *Hypocreales* clone Swis_4.5_122_76 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      32  .....   51

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      393 C.....   370
```

>[NR_175069.1](#) *Fusarium iranicum* CBS 143608 ITS region; from TYPE material

```
product length = 355
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      226 .....   245

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      580 C.....   557
```

>[KC808249.1](#) *Fusarium petroliphilum* culture NRRL:54985 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 366
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      186 .....   205

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      551 C.....   528
```

>[MK406566.1](#) Uncultured *Fusarium* clone D762ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 233 ..... 252

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 586 C..... 563

```

>[JX220225.1](#) *Fusarium* sp. D/Sv/1/4 18S ribosomal RNA, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA, partial sequence

```

product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 143 ..... 162

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 484 C..... 461

```

>[EU750678.1](#) *Fusarium* sp. 08025 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 321 ..... 340

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 681 C..... 658

```

>[MK406405.1](#) Uncultured *Fusarium* clone D601ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 233 ..... 252

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 587 C..... 564

```

>[OM735578.1](#) *Sarocladium zeae* voucher OH00004_JVE-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 95 ..... 114

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 453 C..... 430

```

>[OM515069.1](#) *Trichoderma koningiopsis* strain CEN203 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 365

```

```

Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      333  ..... 352

Reverse primer 1    NTTTCCTCCGCTTATTGATATGC  24
Template      697  C..... 674

```

>[MN954341.1](#) *Fusarium variasi* isolate CMW54696 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      199  ..... 218

Reverse primer 1    NTTTCCTCCGCTTATTGATATGC  24
Template      559  C..... 536

```

>[OM837272.1](#) *Fusarium verticillioides* culture NFCCI:2696 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      28   ..... 47

Reverse primer 1    NTTTCCTCCGCTTATTGATATGC  24
Template      370  C..... 347

```

>[HE649457.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT22

```

product length = 362
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      211  ..... 230

Reverse primer 1    NTTTCCTCCGCTTATTGATATGC  24
Template      572  C..... 549

```

>[PV335618.1](#) *Fusarium solani* isolate Fs5 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      14   ..... 33

Reverse primer 1    NTTTCCTCCGCTTATTGATATGC  24
Template      374  C..... 351

```

>[MK406608.1](#) Uncultured *Fusarium* clone D804ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1    GCATCGATGAAGAACGCAGC  20
Template      233  ..... 252

Reverse primer 1    NTTTCCTCCGCTTATTGATATGC  24

```

Template 586 C..... 563

>[OQ955751.1](#) *Fusarium solani* isolate F10 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 180 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

>[HE649431.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT62

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 208 227

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 569 C..... 546

>[LC681745.1](#) *Claviceps queenslandica* MAFF 306124 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 379

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 214 233

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 592 C..... 569

>[OM570550.1](#) *Fusarium tricinctum* strain FUP01 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 72 91

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 427 C..... 404

>[OL832190.1](#) *Fusarium* sp. FTSC_11 strain FRC R6763 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 173 192

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 527 C..... 504

>**MH862670.1** *Fusarium napiforme* culture CBS:748.97 strain CBS 748.97 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 319 338

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 661 C..... 638

>**MK406593.1** Uncultured *Clonostachys* clone D789ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 256 275

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 612 C..... 589

>**PX312713.1** *Fusarium incarnatum* isolate HvrMPb13 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 564 C..... 541

>**OM106679.1** *Fusarium* sp. isolate GLBRC483 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 524 C..... 501

>**MH333109.1** *Fusarium oxysporum* isolate Ya5 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 227 246

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 580 C..... 557

>[KC007133.1](#) *Neonectria* sp. BESC103d internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       171    ..... 190

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       522    C..... 499
```

>[AB100361.1](#) *Nomuraea rileyi* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```
product length = 381
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       1958    ..... 1977

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       2337    -..... 2315
```

>[MH284717.1](#) *Trichoderma* sp. isolate yi0459_1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 363
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       232    ..... 251

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       593    -..... 571
```

>[OM515008.1](#) *Trichoderma afroharzianum* strain CEN1328 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       351    ..... 370

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       712    C..... 689
```

>[ON931729.1](#) *Fusarium* sp. strain KOD 1271 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 348
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       234    ..... 253

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       581    C..... 558
```

>[HQ591384.1](#) *Cordyceps militaris* strain BCRC33735 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2,

complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 351
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 408 ..... 427

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 758 C..... 735
```

>[HF546283.1](#) *Geosmithia morbida* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U173

```
product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 219 ..... 238

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 583 C..... 560
```

>[OQ513884.1](#) *Metapochonia rubescens* strain DTO_461-A2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 371
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 349 ..... 368

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 719 C..... 696
```

>[JN235241.1](#) *Fusarium petroliphilum* strain FRC S-2481 5.8S ribosomal RNA gene, complete sequence

```
product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 ..... 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521
```

>[MZ742820.1](#) *Fusarium coicis* isolate DG11022037 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 343
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 257 ..... 276

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 599 C..... 576
```

>[KF428546.1](#) *Neonectria* sp. nc_ca_0101c internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 352
```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       176    ..... 195

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       527    C..... 504

```

>[AB255606.1](#) *Tolypocladium inflatum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: NBRC 31669

```

product length = 343
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       2352    ..... 2371

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       2694    C..... 2671

```

>[OR355729.1](#) *Trichoderma* sp. isolate P17 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       235    ..... 254

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       597    C..... 574

```

>[JQ411382.1](#) *Clonostachys rosea* isolate C40254P301CC519 internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       176    ..... 195

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       532    C..... 509

```

>[OU720034.1](#) *Claviceps sulcata* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CCC1460

```

product length = 376
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       168    ..... 187

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       543    C..... 520

```

>[OM515059.1](#) *Trichoderma ghanense* strain CEN555 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 365
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       368    ..... 387

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24

```

Template 732 C..... 709

>[JF303873.1](#) *Hypocrea longipilosa* strain TL1 18S ribosomal RNA gene, internal transcribed spacer 1, 5.8S ribosomal RNA gene, internal transcribed spacer 2, and 28S ribosomal RNA gene, region

product length = 358

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 204 223

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 561 C..... 538

>[AB498983.1](#) *Fusarium solani* f. sp. *eumartii* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: MAFF 240345

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 213 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 575 C..... 552

>[ON931759.1](#) *Fusarium decemcellulare* strain F167 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 347

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 234 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 580 C..... 557

>[AB255605.1](#) *Claviceps panicoidearum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: MAFF 511351

product length = 380

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 2327 2346

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 2705 -..... 2683

>[AM181458.2](#) *Geosmithia* sp. MK1683a 18S rRNA gene (partial), 5.8S rRNA gene, 28S rRNA gene (partial), ITS1 and ITS2, strain MK1683a

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 576 C..... 553

>[GQ505763.1](#) *Fusarium concolor* strain NRRL 13459 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 343
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 222 ..... 241

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 564 C..... 541

```

>[JX238537.1](#) *Fusarium* sp. W/Asd/11/1 18S ribosomal RNA, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 153 ..... 172

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 493 C..... 470

```

>[KU504281.1](#) *Fusarium* sp. BAB-3272 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 209 ..... 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 570 C..... 547

```

>[MG274305.1](#) *Fusarium equiseti* isolate 05005 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 224 ..... 243

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 565 C..... 542

```

>[MH795087.1](#) *Sphaerostilbella broomeana* voucher TFC:200254 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 372
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 223 ..... 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 594 C..... 571

```

>[JQ766113.1](#) *Metarhizium anisopliae* strain TSBP1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367

```

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 200 ..... 219

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 565 -..... 543
```

>[PQ634817.1](#) *Fusarium* sp. isolate Ab.225 TH small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 340
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 478 ..... 497

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 817 C..... 794
```

>[MT557329.1](#) *Trichoderma* sp. isolate RL647 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 198 ..... 217

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 559 C..... 536
```

>[AJ875333.1](#) *Neonectria radiculicola* 5.8S rRNA gene, ITS1 and ITS2, isolate cy23

```
product length = 353
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 207 ..... 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 559 C..... 536
```

>[LT216452.1](#) *Claviceps cynodontis* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCC 651

```
product length = 377
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 545 C..... 522
```

>[ON931737.1](#) *Fusarium* sp. strain NRRL 22511 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 347
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 234 ..... 253

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 580 C..... 557
```

>[KC461476.1](#) Uncultured Hypocreales clone Swis_1.3_24_10 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 358

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	32		51
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	389	C.....	366

>[NR_169922.1](#) *Fusarium tenuicristatum* IMI 277708 ITS region; from TYPE material

product length = 356

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	226		245
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	581	C.....	558

>[PX350784.1](#) *Fusarium* sp. isolate LLC3042 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	221		240
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	563	C.....	540

>[FJ765008.1](#) *Isaria fumosorosea* isolate CNZH 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	255		274
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	606	C.....	583

>[KM231835.1](#) *Allantonectria miltina* strain CBS 474.69 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 353

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	225		244
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	577	C.....	554

>[PV593241.1](#) *Trichoderma* sp. isolate GLBRC1477 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 358
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 254 ..... 273

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 611 C..... 588

```

>[MN644521.1](#) *Trichoderma effusum* voucher research collection Farrer lab 233 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 346
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 227 ..... 246

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 572 C..... 549

```

>[EF453133.1](#) *Fusarium* sp. NRRL 43650 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 532 C..... 509

```

>[OL832310.1](#) *Fusarium flocciferum* strain NRRL 45999 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 ..... 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515

```

>[KM231800.1](#) [*Neocosmospora*] *rubicola* strain CBS 101018 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 231 ..... 250

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 591 C..... 568

```

>[MH825715.1](#) *Trichoderma asperellum* isolate IIRCK2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 364

```

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      180      ..... 199

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      543      C..... 520

```

>[MW989734.1](#) *Fusarium falciforme* strain JUS1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      189      ..... 208

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      550      C..... 527

```

>[PV593206.1](#) *Metarhizium* sp. isolate GLBRC1434 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 369
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      258      ..... 277

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      626      C..... 603

```

>[KT878339.1](#) *Acremonium acutatum* strain 03914 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 372
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      265      ..... 284

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      636      C..... 613

```

>[PQ326619.1](#) *Gliocephalotrichum simplex* strain MFLUCC 24-0051 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      315      ..... 334

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      674      C..... 651

```

>[FJ430783.1](#) *Trichoderma koningii* strain CCF3834 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 365
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      232      ..... 251

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 596 C..... 573

>[GQ121305.1](#) *Fusarium oxysporum* isolate GIFUDHAN8 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 341
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 178 197
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 517 -..... 495

>[HQ130672.1](#) *Fusarium* sp. WF113 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 207 226
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 548 C..... 525

>[HF546288.1](#) *Geosmithia* sp. U197 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U197

product length = 365
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 587 C..... 564

>[OM106600.1](#) *Fusarium* sp. isolate GLBRC403 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 207 226
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 561 C..... 538

>[EU307906.1](#) *Metarhizium anisopliae* var. *anisopliae* strain LRC 207 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 571 590
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24

Template 937 C..... 914

>[OP510082.1](#) *Fusarium* sp. isolate B8.Contig_consensus_sequence internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 522 C..... 499

>[PX350198.1](#) *Fusarium* sp. isolate LLC1697 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 231 250

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 584 C..... 561

>[MH550509.1](#) *Ijuhya* sp. strain daef39 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 382 401

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 733 C..... 710

>[OK298423.1](#) [*Gibberella*] *fujikuroi* var. *moniliformis* isolate OJ18 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 198 217

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

>[MK280794.1](#) *Fusarium nanum* strain LC12168 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 160 179

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 501 C..... 478

>[OM423350.1](#) *Purpureocillium lilacinum* isolate GZUIFR 22.153 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       221    ..... 240

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       572    C..... 549
```

>[HG936544.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09S70C85 (MOTU42)

```
product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       251    ..... 270

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       602    C..... 579
```

>[LT746248.1](#) *Fusarium ensiforme* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CPC 27191

```
product length = 361
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       196    ..... 215

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       556    C..... 533
```

>[HG936113.1](#) Uncultured *Fusidium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09S50C87 (MOTU18)

```
product length = 364
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       249    ..... 268

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       612    C..... 589
```

>[AY188917.1](#) *Fusarium sporotrichioides* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       210    ..... 229

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       550    C..... 527
```

>[PP125840.1](#) *Trichoderma* sp. isolate LCMP-T3 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 215 ..... 234

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 578 C..... 555

```

>[HQ115655.1](#) *Trichothecium roseum* isolate E04 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 384
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 202 ..... 221

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

```

>[MK407959.1](#) Uncultured *Fusarium* clone D2190ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 232 ..... 251

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

```

>[MZ423771.1](#) *Nectriaceae* sp. isolate GW20_567 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 170 ..... 189

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 521 C..... 498

```

>[KR906695.1](#) *Fusarium solani* isolate U5 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 146 ..... 165

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 507 C..... 484

```

>[OM106272.1](#) *Fusarium* sp. isolate GLBRC71 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341

```

```
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 160 ..... 179

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 500 C..... 477
```

>[AJ311951.2](#) *Claviceps purpurea* 18S rRNA gene (partial), 5.8S rRNA gene, 28S rRNA gene (partial), internal transcribed spacer 1 (ITS1) and internal transcribed spacer 2 (ITS2), isolate 511

```
product length = 370
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 213 ..... 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 582 C..... 559
```

>[JX243930.1](#) *Neonectria* sp. 1n internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 172 ..... 191

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 523 C..... 500
```

>[OQ421962.1](#) *Fusarium* sp. breed MHUB11357 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 570 T..... 547
```

>[ON207922.1](#) *Fusarium boothii* strain CN062C4 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 317 ..... 336

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 658 C..... 635
```

>[LC681715.1](#) *Claviceps litoralis* TNS:F-96439 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```
product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 281 ..... 300

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
```

Template 643 C..... 620

>[HF546272.1](#) *Geosmithia fassatae* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U140

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 586 C..... 563

>[NR_163507.1](#) *Claviceps loudetiae* CCC 656 ITS region; from TYPE material

product length = 375

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 211 230

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

>[MZ433197.1](#) *Clonostachys rosea* isolate HQ-2-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 147 166

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 503 C..... 480

>[LC793072.1](#) *Lecanicillium* sp. CKSI15-F14-E1-i4 genes for ITS1, 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 229 248

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 582 C..... 559

>[KU837842.1](#) *Epichloe* sp. strain A168I internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 216 235

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 583 C..... 560

>[KF428572.1](#) *Neonectria* sp. nwa_besc_15a internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       175 .....   194

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       526 C.....   503

```

>[EU091009.1](#) *Fusarium oxysporum* isolate F35 internal transcribed space 1 gene, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       175 .....   194

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       515 C.....   492

```

>[LR583781.1](#) *Fusarium bataticola*

```

product length = 337
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       236 .....   255

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       572 C.....   549

```

>[OM846598.1](#) *Trichoderma spirale* isolate CNUFC-RD150 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 358
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       234 .....   253

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       591 C.....   568

```

>[KC007323.1](#) *Neonectria* sp. O_2_BESC_883h internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       169 .....   188

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       522 C.....   499

```

>[LT746276.1](#) *Neocosmospora* sp. FSSC 28 genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CPC 28194

```

product length = 361
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       205 .....   224

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24

```

Template 565 C..... 542

>[EU091039.1](#) *Fusarium proliferatum* isolate F25 internal transcribed space 1 gene, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 531 C..... 508

>[AF210671.1](#) *Stephanonectria keithii* strain CBS 100005 internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 193 212

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 553 C..... 530

>[AJ605994.2](#) *Claviceps africana* var. *hyparrheniae* 18S rRNA gene (partial), 5.8S rRNA gene, 28S rRNA gene (partial), ITS1 and ITS2, isolate CCC 570

product length = 382

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 593 C..... 570

>[KT347160.1](#) *Fusarium* sp. isolate FEV8B internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 562 C..... 539

>[PP386584.1](#) *Trichoderma* sp. voucher 47S small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 1988 2007

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 2350 C..... 2327

>[OL832236.1](#) *Fusarium acuminatum* strain FRC R8222 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 540 C..... 517

>[PQ340456.1](#) *Fusarium sporotrichioides* isolate ITS185 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 546 C..... 523

>[KC244316.1](#) *Metacordyceps taii* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 2726 2745

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 3092 C..... 3069

>[HF546226.1](#) *Geosmithia* sp. CCF3641 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF3641, isolate U119b

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 213 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 578 C..... 555

>[MZ425965.1](#) *Geosmithia morbida* isolate GmA internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 156 175

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 520 C..... 497

>[OM106442.1](#) *Fusarium* sp. isolate GLBRC244 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer

2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 551 C..... 528

>[MT366772.1](#) *Fusarium* sp. isolate XXMF000008 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 209 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 563 T..... 540

>[MH300451.1](#) *Fusarium solani* isolate MAF internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 189 208

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 549 C..... 526

>[AJ557074.3](#) *Claviceps setariiphila* 18S rRNA gene (partial), ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene (partial), isolate CCC 405

product length = 384

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 175 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 558 C..... 535

>[JF303879.1](#) *Hypocrea viridescens* strain HV3 18S ribosomal RNA gene, internal transcribed spacer 1, 5.8S ribosomal RNA gene, internal transcribed spacer 2, and 28S ribosomal RNA gene, region

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 195 214

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 559 T..... 536

>[EU871033.1](#) *Hypocrea lixii* strain TU Graz 11TSM3 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 362

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       242    ..... 261

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       603    C..... 580

```

>[JX243927.1](#) *Neonectria* sp. 1j internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       172    ..... 191

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       523    C..... 500

```

>[PP947375.1](#) *Fusarium poae* strain CN104E9 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       310    ..... 329

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       650    C..... 627

```

>[OR654961.1](#) *Fusicolla violacea* strain KUC21850 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       153    ..... 172

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       507    C..... 484

```

>[JN235250.1](#) *Fusarium petroliphilum* strain FRC S-2506 5.8S ribosomal RNA gene, complete sequence

```

product length = 364
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       166    ..... 185

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       529    C..... 506

```

>[JF740894.1](#) *Fusarium* sp. NRRL 25128 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       169    ..... 188

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 523 C..... 500

>[KC808252.1](#) *Fusarium petroliphilum* culture NRRL:54988 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 549 C..... 526

>[PQ340452.1](#) *Fusarium sporotrichioides* isolate ITS24 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 199 218
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 539 C..... 516

>[MF363050.1](#) *Trichoderma asperellum* strain M103 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 364
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 243 262
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 606 C..... 583

>[HE974455.1](#) *Fusarium solani* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF 4358

product length = 362
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 197 216
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 558 C..... 535

>[LT216449.1](#) *Claviceps sorghi* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCC 1028

product length = 375
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 221 240
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 595 C..... 572

>[KT347152.1](#) *Fusarium* sp. isolate FEB5C internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       204    ..... 223

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       544    C..... 521
```

>[JQ606830.1](#) *Trichothecium crotocinigenum* strain NRRL 53579 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 369
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       199    ..... 218

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       567    C..... 544
```

>[LC728645.1](#) *Claviceps imperatae* MAFF 247731 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```
product length = 381
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       241    ..... 260

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       621    C..... 598
```

>[AJ605996.2](#) *Claviceps* sp. CCC 647 18S rRNA gene (partial), ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene (partial), isolate CCC 647

```
product length = 371
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       221    ..... 240

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       591    C..... 568
```

>[MW298995.1](#) *Fusarium* sp. isolate 836504-17-2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       136    ..... 155

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       490    C..... 467
```

>[MN494089.1](#) *Beauveria bassiana* strain SZY1 18S ribosomal RNA gene, internal transcribed spacer 1, 5.8S ribosomal RNA gene, internal transcribed spacer 2, 28S ribosomal RNA gene, and 28S-18S ribosomal RNA intergenic spacer, complete sequence

```

product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       1988  .....  2007

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       2339  C.....  2316

```

>[PX313668.1](#) *Fusarium solani* isolate F2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       221    .....  240

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       581    C.....  558

```

>[MT557142.1](#) *Fusarium* sp. isolate RL833 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       146    .....  165

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       500    C.....  477

```

>[KJ542162.1](#) *Metarhizium robertsii* isolate M329-08 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       138    .....  157

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       504    C.....  481

```

>[LC793063.1](#) *Lecanicillium* sp. CKSI15-F13-TL-i3 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```

product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       209    .....  228

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template       563    C.....  540

```

>[MN882831.1](#) *Fusarium chlamydosporum* strain DTO 418-C1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1      GCATCGATGAAGAACGCAGC  20

```

Template	305		324
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	661	C.....	638

>[ON411502.1](#) *Fusarium nygamai* isolate T1.17 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354			
Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	159		178
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	512	C.....	489

>[JN235546.1](#) *Fusarium* cf. *dimerum* a DPGS-2011 strain FRC E-334 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359			
Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	126		145
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	484	C.....	461

>[OP346773.1](#) *Clonostachys rosea* isolate EEK_D3 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357			
Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	332		351
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	688	C.....	665

>[FJ430712.1](#) *Emericellopsis fimetaria* strain MH178 18S ribosomal RNA gene and internal transcribed spacer 1, partial sequence; and 5.8S ribosomal RNA gene, internal transcribed spacer 2, and 28S ribosomal RNA gene, complete sequence

product length = 360			
Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	1999		2018
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	2358	C.....	2335

>[JX435205.1](#) *Fusarium* cf. *solani* CBS 2012 strain CBS 117481 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361			
Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	180		199

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 540 C..... 517

>[OM106461.1](#) *Fusarium* sp. isolate GLBRC263 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 216 235
 Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 556 C..... 533

>[KC007326.1](#) *Neonectria* sp. O_2_BESC_883k internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 168 187
 Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 521 C..... 498

>[GQ167228.1](#) *Sarocladium zeae* culture NRRL:47823 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 359
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 197 216
 Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 555 C..... 532

>[MW791972.1](#) *Fusarium oxysporum* voucher culture Y324N internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 185 204
 Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 525 C..... 502

>[OM106342.1](#) *Clonostachys* sp. isolate GLBRC144 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 212 231
 Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24

Template 568 C..... 545

>[MT435069.1](#) *Fusarium spartum* strain NRRL_66895_F_spartum internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 183 202

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 537 C..... 514

>[HF546368.1](#) *Gibberella moniliformis* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain TUEF9

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 158 177

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 511 C..... 488

>[OM515072.1](#) *Trichoderma koningiopsis* strain CEN980 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 333 352

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 697 C..... 674

>[LC702056.1](#) *Fusarium populicola* DF02 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, and 28S rRNA

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 185 204

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521

>[KY426420.1](#) *Fusarium graminearum* isolate HS27 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 517 -..... 495

>[OQ434140.1](#) *Fusarium* sp. isolate NB134 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit

ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 176 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 530 C..... 507

>[HF680221.1](#) *Acremonium cavaeae* genomic DNA containing ITS1, 5.8S rRNA gene and ITS2, strain CBS 111656

product length = 369

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 550 C..... 527

>[MH550479.1](#) *Fusarium* sp. strain daef9 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 376 395

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 731 C..... 708

>[JN943378.1](#) *Hypocrea sulphurea* strain NBRC 8438 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 228 247

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 592 C..... 569

>[GU055584.1](#) Uncultured *Hypocreales* clone NG_N_E11 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 581 C..... 558

>[OM106592.1](#) *Fusarium* sp. isolate GLBRC395 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 563 C..... 540

```

>[FJ545285.1](#) *Metarhizium anisopliae* isolate CNGD10 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 547 ..... 566

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 913 C..... 890

```

>[HF546281.1](#) *Geosmithia pallida* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U17

```

product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 217 ..... 236

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 581 C..... 558

```

>[HG936108.1](#) Uncultured *Fusidium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09S50C57 (MOTU18)

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 249 ..... 268

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 612 C..... 589

```

>[OM106371.1](#) *Fusarium* sp. isolate GLBRC173 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 200 ..... 219

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

```

>[KC461467.1](#) Uncultured *Hypocreales* clone Rhir_4.4_120_35 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 32 ..... 51

```

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 390 C..... 367

>[KT876651.1](#) *Fusarium oxysporum* isolate C9-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 164 183

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 504 C..... 481

>[EF060487.1](#) *Hypocreales* sp. LM120 18S ribosomal RNA gene, partial sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 535 554

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 885 T..... 862

>[MH550505.1](#) *Trichoderma* sp. strain daef35 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 405 424

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 766 C..... 743

>[EU214563.1](#) *Fusarium oxysporum* f. sp. *ciceris* NBAIM:2791 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 327

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 173 192

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 499 C..... 476

>[OL832181.1](#) *Fusarium reticulatum* var. *negundinis* strain NRRL 20682 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 180 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 535 C..... 512

>[EF052277.1](#) *Sphacelia texensis* isolate CCC 774 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 378
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      220    ..... 239

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      597    C..... 574
```

>[AB378514.1](#) *Lecanicillium attenuatum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: KYK00326

```
product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      243    ..... 262

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      594    C..... 571
```

>[MW298958.1](#) *Fusarium* sp. isolate 836489-15 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      141    ..... 160

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      495    C..... 472
```

>[PX350192.1](#) *Fusarium* cf. *solani* isolate LLC1690 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      226    ..... 245

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      587    C..... 564
```

>[HE649440.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT76

```
product length = 363
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      209    ..... 228

Reverse primer 1      NTTTTCTCCGCTTATTGATATGC  24
Template      571    C..... 548
```

>[JN235263.1](#) *Fusarium petroliphilum* strain FRC S-2546 5.8S ribosomal RNA gene, complete sequence

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 ..... 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521

```

>[MT078888.1](#) *Metarhizium brunneum* strain CBS 316.51 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 313 ..... 332

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 679 C..... 656

```

>[KU872831.1](#) *Fusarium oxysporum* isolate AFIC26 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA region gene, partial sequence

```

product length = 356
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 322 ..... 341

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 677 C..... 654

```

>[KC427079.1](#) *Stachybotrys chartarum* strain F277954 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 366
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 199 ..... 218

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 564 C..... 541

```

>[MZ089667.1](#) *Fusarium pseudograminearum* isolate Fs-B24 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 339
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 178 ..... 197

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 515 -..... 493

```

>[HG936817.1](#) Uncultured *Neonectria* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09S10C47 (MOTU78)

```

product length = 353
Forward primer 1 GCATCGATGAAGAACGCAGC 20

```

```

Template      234 ..... 253

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      586 C..... 563

```

>[KC461453.1](#) Uncultured Hypocreales clone Epol_4.4_121_2 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      32 ..... 51

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      390 C..... 367

```

>[KM231834.1](#) Xenoacremonium recifei strain CBS 541.89 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 353
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      228 ..... 247

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      580 C..... 557

```

>[KJ542165.1](#) Metarhizium brunneum isolate M002-08 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      145 ..... 164

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      511 C..... 488

```

>[MW789356.1](#) Fusarium proliferatum isolate E8-2018 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      358 ..... 377

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      719 C..... 696

```

>[KF723006.1](#) Myrothecium sp. W MG-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 357
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      315 ..... 334

```

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 671 C..... 648

>[KT876641.1](#) *Fusarium solani* isolate C10-4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 172 191

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 533 C..... 510

>[MT071407.1](#) *Fusarium acuminatum* isolate XRXBK internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 186 205

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 540 T..... 517

>[EU307924.1](#) *Metarhizium anisopliae* var. *anisopliae* isolate P3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 367

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 532 551

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 898 C..... 875

>[AJ243778.1](#) *Cordyceps sinensis* partial 18S rRNA gene, 5.8S rRNA gene, partial 26S rRNA gene and ITS1 and 2, stroma

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 192 211

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 554 A..... 531

>[KM051389.1](#) *Trichoderma* sp. BAB-3898 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 364

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 241 260

Reverse primer 1 NT TTCCTCCGCTTATTGATATGC 24
 Template 604 C..... 581

>[MN644706.1](#) *Hypocreales* sp. voucher research collection Farrer lab 52 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 343
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       232      ..... 251

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       574      C..... 551
```

>[KR906697.1](#) *Fusarium solani* isolate U7.1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       107      ..... 126

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       468      C..... 445
```

>[PV640836.1](#) *Fusarium culmorum* isolate Karbala-1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       592      ..... 611

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       933      C..... 910
```

>[FJ765016.1](#) *Isaria fumosorosea* isolate NLAG 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       258      ..... 277

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       609      C..... 586
```

>[AJ301993.1](#) *Myrothecium roridum* 18S rRNA gene, 5.8S rRNA gene, 28S rRNA gene (partial), internal transcribed spacer 1 (ITS1) and internal transcribed spacer 2 (ITS2), strain BBA 62764

```
product length = 381
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       1971      ..... 1990

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       2351      C..... 2328
```

>[PV593209.1](#) *Fusarium* sp. isolate GLBRC1438 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit

ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 199 218

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 539 C..... 516

>[KU837765.1](#) *Epichloe* sp. strain A255C internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 207 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 574 C..... 551

>[OM106817.1](#) *Fusarium* sp. isolate GLBRC630 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 176 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 516 C..... 493

>[GU055518.1](#) Uncultured *Myrothecium* clone NG_M_A01 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 241 260

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 602 C..... 579

>[PP336466.1](#) *Trichoderma longibrachiatum* strain MK1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 398 417

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 756 C..... 733

>[XR_001936381.1](#) *Fusarium oxysporum* f. sp. *lycopersici* 4287 28S ribosomal RNA (FOXG_19919), partial rRNA

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 29 ..... 48

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 369 C..... 346

```

>[LC793066.1](#) *Lecanicillium* sp. CKSI15-F13-TR-i2 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 564 C..... 541

```

>[AB752247.1](#) *Nectriaceae* sp. AS4-1 genes for ITS1, 5.8S rRNA, ITS2, and 28S rRNA

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 194 ..... 213

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 547 C..... 524

```

>[MW299461.1](#) *Hypocreales* sp. isolate 928717-37 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 139 ..... 158

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 499 C..... 476

```

>[JX083509.1](#) *Claviceps humidiphila* isolate CCC981 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 369
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 213 ..... 232

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 581 C..... 558

```

>[KY484941.1](#) *Fusarium solani* isolate Fs-54S internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 186 ..... 205

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24

```

Template 547 C..... 524

>[MK280798.1](#) *Fusarium ipomoeae* strain LC7940 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 160 179

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 501 C..... 478

>[KC311517.1](#) *Fusarium equiseti* isolate SC1104_10 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 212 231

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 553 C..... 530

>[MG865983.1](#) *Fusarium oxysporum* voucher Fo internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 149 168

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 503 C..... 480

>[OQ509676.1](#) *Fusarium pseudensiforme* strain 1528 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 584 C..... 561

>[LC571760.1](#) *Aciculosporium oplismeni* MAFF:246966 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 527 C..... 504

>[HG936548.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10S50C03 (MOTU42)

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 252 271

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 603 C..... 580

>[KC461308.1](#) Uncultured *Hypocreales* clone Jmex_4.3_115_12 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 410

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 32 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 441 C..... 418

>[HG937015.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10SWuC47 (MOTU147)

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 249 268

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 608 C..... 585

>[LC145810.1](#) *Calcarisporium arbuscula* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: 1310-04

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 187 206

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 546 C..... 523

>[OL832260.1](#) *Fusarium tricinctum* strain FRC T904 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 176 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 530 C..... 507

>[LC681723.1](#) *Claviceps miscanthicola* TNS:F-96458 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 380

```

Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      220  ..... 239

Reverse primer 1      NTTTTCCTCCGCTTATTGATATGC  24
Template      599  C..... 576

```

>[PP345623.1](#) *Fusarium commune* isolate BT2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      180  ..... 199

Reverse primer 1      NTTTTCCTCCGCTTATTGATATGC  24
Template      534  C..... 511

```

>[OP743920.1](#) *Fusarium solani* isolate BJ1-B1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      204  ..... 223

Reverse primer 1      NTTTTCCTCCGCTTATTGATATGC  24
Template      564  C..... 541

```

>[HF569182.1](#) *Hypocrea nigricans* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain FCCTM7

```

product length = 362
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      291  ..... 310

Reverse primer 1      NTTTTCCTCCGCTTATTGATATGC  24
Template      652  G..... 629

```

>[HQ207030.1](#) *Hypocreales* sp. 158_PH internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      144  ..... 163

Reverse primer 1      NTTTTCCTCCGCTTATTGATATGC  24
Template      485  C..... 462

```

>[KT347156.1](#) *Fusarium* sp. isolate FEB107B internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template      198  ..... 217

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 538 C..... 515

>[ON074993.1](#) *Fusarium chlamydosporum* strain CN026D2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 312 331

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 668 C..... 645

>[OM372730.1](#) *Purpureocillium lavendulum* strain PulaHB2102 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 215 234

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 565 T..... 542

>[AJ578485.2](#) *Geosmithia pallida* 18S rRNA gene (partial), 5.8S rRNA gene, 28S rRNA gene (partial), ITS1 and ITS2, isolate CCF 3340

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 208 227

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 572 C..... 549

>[NR_165844.1](#) *Fusarium obliquiseptatum* NRRL 62611 ITS region; from TYPE material

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 154 173

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 513 C..... 490

>[OM106723.1](#) *Fusarium* sp. isolate GLBRC527 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 184 203

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 524 C..... 501

>[KM242329.1](#) *Neonectria* sp. INBio_4506T internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 357

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	32		51
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	388	C.....	365

>[FN859421.1](#) *Cladobotryum* sp. FSU 5046 genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain FSU 5046

product length = 370

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	206		225
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	575	C.....	552

>[OM106594.1](#) *Fusarium* sp. isolate GLBRC397 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	210		229
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	550	C.....	527

>[KF428673.1](#) *Ilyonectria* sp. nwa_sqmc_26_5a internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	146		165
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	500	C.....	477

>[KM231855.1](#) *Calostilbe striispora* strain CBS 122.39 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 372

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	276		295
Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	647	C.....	624

>[KC305345.1](#) *Stachybotrys* sp. CYJ-2013b voucher HGUP 0146 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer

2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 287 306

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 652 C..... 629

>[MN644679.1](#) *Trichoderma afroharzianum* voucher research collection Farrer lab 245 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 594 C..... 571

>[MK828118.1](#) *Fusarium graminearum* strain 140 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 339

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 933 952

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 1271 C..... 1248

>[MZ423560.1](#) *Hypocreales* sp. isolate GW_Cank20_148 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 188 207

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 549 C..... 526

>[KM215643.1](#) *Gibberella* sp. G345 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 211 230

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 565 C..... 542

>[MW298938.1](#) *Fusarium* sp. isolate 836467-37 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        129  .....          148

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        483  C.....          460

```

>[AB378540.1](#) *Simplicillium* sp. ID06-F0394 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```

product length = 380
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        204  .....          223

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        583  C.....          560

```

>[JF519256.1](#) Uncultured *Trichoderma* clone RELIS_K6_A05 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 358
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        276  .....          295

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        633  C.....          610

```

>[FJ179466.1](#) Uncultured *Fusarium* clone GSF1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        233  .....          252

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        586  C.....          563

```

>[AB189443.1](#) *Cordyceps takaomontana* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: NIFTS HF701

```

product length = 349
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        287  .....          306

Reverse primer  1    NTTTTCTCCGCTTATTGATATGC   24
Template        635  C.....          612

```

>[MT228400.1](#) *Acremonium sclerotigenum* strain BAF22 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        1057 .....          1076

```

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 1418 C..... 1395

>[OM106796.1](#) *Fusarium* sp. isolate GLBRC609 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 181 200
 Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 541 C..... 518

>[MZ423777.1](#) *Hypocreaceae* sp. isolate GW20_581 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 365
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 216 235
 Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 580 C..... 557

>[ON193729.1](#) *Gliomastix murorum* strain UNIJAG.PL.OP627 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 369
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 222 241
 Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 590 C..... 567

>[LS479412.1](#) *Fusarium equiseti* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, strain Indo167

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 218 237
 Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 559 C..... 536

>[PX349728.1](#) *Fusarium* sp. isolate LLC919 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 354 373
 Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24

Template 695 C..... 672

>[PV593259.1](#) *Clonostachys rosea* isolate GLBRC1498 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 190 209

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 546 C..... 523

>[HF546349.2](#) *Geosmithia pallida* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U87

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 594 C..... 571

>[EF453072.1](#) *Fusarium* sp. NRRL 43373 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 543 C..... 520

>[NR_189406.1](#) *Niesslia endophytica* DAOM 251709 ITS region; from TYPE material

product length = 358

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 285 304

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 642 C..... 619

>[PX350063.1](#) *Fusarium* sp. isolate LLC1494 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 222 241

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 563 C..... 540

>[HF546345.2](#) *Geosmithia* sp. CCF3355 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF3355, isolate U78

```

product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 ..... 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

```

>[OL832258.1](#) *Fusarium tricinctum* strain NRRL 36381 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 356
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 174 ..... 193

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 529 C..... 506

```

>[JN235316.1](#) *Fusarium petroliphilum* strain FRC S-2504 5.8S ribosomal RNA gene, complete sequence

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 ..... 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 544 C..... 521

```

>[HG935557.1](#) Uncultured *Verticillium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 10D50C63 (MOTU6)

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 240 ..... 259

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 601 C..... 578

```

>[OM515079.1](#) *Trichoderma longibrachiatum* strain CEN1562 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 373 ..... 392

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 732 C..... 709

```

>[JX231154.1](#) *Ilyonectria* sp. LL-2013d culture-collection CBS:132813 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 217 ..... 236

```

Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 570 C..... 547

>[KC007329.1](#) *Lecanicillium* sp. O_3_BESC_246b internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 351
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 567 C..... 544

>[KM491887.1](#) *Trichoderma gamsii* clone 2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 365
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 319 338
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 683 C..... 660

>[OM515012.1](#) *Trichoderma afroharzianum* strain CEN158 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 350 369
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 711 C..... 688

>[JN995623.1](#) *Nectria cucurbitula* strain CBS17873 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 352
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 243 262
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24
 Template 594 C..... 571

>[JX435197.1](#) *Fusarium* cf. *solani* CBS 2012 strain CBS 102824 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 180 199
 Reverse primer 1 NTTCCTCCGCTTATTGATATGC 24

Template 541 C..... 518

>[HE649464.1](#) *Hypocrea lixii* genomic DNA containing 18S ribosomal RNA gene, ITS1, 5.8S ribosomal RNA gene, and ITS2, and 28S ribosomal RNA gene, isolate TUT37

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 568 C..... 545

>[KP143703.1](#) *Sarocladium implicatum* strain B1aA1PD2CC688 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 186 205

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 551 C..... 528

>[AB586996.1](#) *Fusarium dimerum* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: MAFF 237465

product length = 356

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 185 204

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

>[ON493028.1](#) *Trichoderma chlorosporum* strain SYN802 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 335

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 229 248

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 563 C..... 540

>[KF367486.1](#) *Trichoderma* sp. 6 BRO-2013 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 333 352

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 695 C..... 672

>[OM106688.1](#) *Fusarium* sp. isolate GLBRC492 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 193 212

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 533 C..... 510

>[NR_161154.1](#) *Lasionectria hilhorstii* CBS 144938 ITS region; from TYPE material

product length = 368

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 597 C..... 574

>[OM106457.1](#) *Myrothecium* sp. isolate GLBRC259 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 210 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 561 C..... 538

>[MT453297.1](#) *Metarhizium marquandii* isolate DSM100410_J_RLCS24 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 375

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 269 288

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 643 C..... 620

>[KX681610.1](#) *Fusarium falciforme* isolate H06A_4L_2b internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 125 144

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 486 C..... 463

>[HF546290.1](#) *Geosmithia pallida* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U2

```

product length = 365
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      226 .....   245

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      590 C.....   567

```

>[OL832293.1](#) *Fusarium iranicum* strain FRC R9551 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      184 .....   203

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      538 C.....   515

```

>[NR_121208.1](#) *Lecanicillium araneicola* NBRC 105407 ITS region; from TYPE material

```

product length = 352
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      252 .....   271

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      603 C.....   580

```

>[HG936854.1](#) Uncultured *Trichoderma* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09SWuC29 (MOTU86)

```

product length = 365
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      283 .....   302

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      647 C.....   624

```

>[OM735576.1](#) *Sarocladium zeae* voucher IL00112_C390-4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      95 .....   114

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template      454 C.....   431

```

>[OM515092.1](#) *Trichoderma rifaii* strain CEN240 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template      349 .....   368

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24

```

Template 710 C..... 687

>[AF461745.1](#) *Paecilomyces fumosoroseus* strain Pfr INRA 35 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 349

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 266 285

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 614 C..... 591

>[HM534894.1](#) *Nectria dematiosa* strain CBS 127383 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 341 360

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 700 C..... 677

>[LC589716.1](#) *Aciculosporium sasicola* MAFF 247297 genes for ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 527 C..... 504

>[MZ423749.1](#) *Nectriaceae* sp. isolate GW20_488 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 542 C..... 519

>[MT557135.1](#) *Fusarium* sp. isolate RL18 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 153 172

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 494 C..... 471

>[OM106501.1](#) *Fusarium* sp. isolate GLBRC303 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 202 221

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 542 C..... 519

>[HG936656.1](#) Uncultured *Fusarium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09D70C44 (MOTU53)

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 246 265

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 587 C..... 564

>[MK407946.1](#) Uncultured *Fusarium* clone D2177ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 232 251

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 585 C..... 562

>[AF210677.1](#) *Clonostachys divergens* strain CBS 967.73b internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 360

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 188 207

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 547 C..... 524

>[JX243950.1](#) *Neonectria* sp. 13f internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 172 191

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 523 C..... 500

>[KX066044.1](#) *Fusarium graminearum* strain JCP64 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2,

complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 197 ..... 216

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 555 C..... 532
```

>[MW299244.1](#) *Fusarium* sp. isolate 880334-34 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 130 ..... 149

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 484 C..... 461
```

>[MZ425505.1](#) *Clonostachys rosea* isolate DG-2-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 357
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 148 ..... 167

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 504 C..... 481
```

>[ON209610.1](#) *Sarocladium strictum* strain CN082D1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 365
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 327 ..... 346

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 691 C..... 668
```

>[OL832326.1](#) *Fusarium nurrugi* strain NRRL 36452 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 358
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 ..... 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 541 C..... 518
```

>[HG936107.1](#) Uncultured *Fusidium* genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, clone 09DWuC57 (MOTU18)

```

product length = 364
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 249 ..... 268

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 612 C..... 589

```

>[PP083207.1](#) *Simplicillium* sp. isolate GCF small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 372
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 252 ..... 271

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 623 T..... 600

```

>[MH873577.1](#) *Lecanicillium antillanum* culture CBS:350.85 strain CBS 350.85 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 356
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 362 ..... 381

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 717 C..... 694

```

>[OL832294.1](#) *Fusarium* sp. FTSC_25 strain FRC R4587 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 356
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 185 ..... 204

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 540 C..... 517

```

>[HQ674657.1](#) *Fusarium* sp. RGT-S4 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 337 ..... 356

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 678 C..... 655

```

>[HQ130709.1](#) *Fusarium* sp. WF153 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 341
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 203 ..... 222

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 543 C..... 520

```

>[EF052282.1](#) *Sphacelia lovelessii* isolate CCC 642 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 370
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 223 ..... 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 592 C..... 569

```

>[MN626668.1](#) *Fusarium* sp. isolate Fus1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 207 ..... 226

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 548 C..... 525

```

>[ON411520.1](#) *Fusarium proliferatum* isolate T7.4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 ..... 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 531 C..... 508

```

>[AB027383.1](#) *Metarhizium anisopliae* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```

product length = 367
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 382 ..... 401

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 748 C..... 725

```

>[MK406589.1](#) Uncultured *Fusarium* clone D785ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20

```

```

Template      233 ..... 252

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      586 C..... 563

```

>[FN859445.1](#) *Hypomyces samuelsii* genomic DNA containing ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain C.L.L. 7259

```

product length = 371
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      206 ..... 225

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      576 C..... 553

```

>[PX244854.1](#) *Subulphorum camptosporum* isolate EG-WSF16 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 368
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      215 ..... 234

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      582 T..... 559

```

>[MN154167.1](#) *Fusarium culmorum* isolate SNO tree 11 leaf 1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      202 ..... 221

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      543 C..... 520

```

>[AF081477.2](#) *Stachybotrys kampalensis* 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 372
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      543 ..... 562

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      914 C..... 891

```

>[KT876652.1](#) *Fusarium oxysporum* isolate C5-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

```

product length = 341
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      165 ..... 184

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24

```

Template 505 C..... 482

>**OQ155178.1** *Fusarium oxysporum* isolate HS12 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 516 -..... 494

>**KC461387.1** Uncultured *Hypocreales* clone Dasp_1.3_22_3 5.8S ribosomal RNA gene, partial sequence; internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 32 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 393 C..... 370

>**MH299308.1** *Sarocladium zeae* voucher CPO 10.325 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 359

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 161 180

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 519 C..... 496

>**JN235252.1** *Fusarium petroliphilum* strain FRC S-2510 5.8S ribosomal RNA gene, complete sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 181 200

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 542 C..... 519

>**GU055602.1** Uncultured *Hypocreales* clone NG_N_H01 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 353

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 230 249

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 582 C..... 559

>[KM816775.1](#) *Hypocreales* sp. G151 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 351
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 173 ..... 192

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 523 C..... 500
```

>[LC792936.1](#) *Lecanicillium* sp. CKSI15-F04-TL-i3 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```
product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 197 ..... 216

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 551 C..... 528
```

>[JX243795.1](#) *Neonectria* sp. NCp07/04 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 176 ..... 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 527 C..... 504
```

>[KJ570695.1](#) *Thyronectria lamyi* strain NL3 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 355 ..... 374

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 709 C..... 686
```

>[AB587005.1](#) *Fusarium mori* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: MAFF 840045

```
product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 184 ..... 203

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 C..... 515
```

>[AJ011783.2](#) *Claviceps africana* ITS1, 5.8S rRNA gene ITS2 and 28S rRNA gene (partial), strain T10 765

```

product length = 383
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        212  .....                231

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        594  C.....                571

```

>[GU055590.1](#) Uncultured Hypocreales clone NG_N_F06 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        230  .....                249

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        581  C.....                558

```

>[NR_154753.1](#) Claviceps langdonii CCC 820 ITS region; from TYPE material

```

product length = 378
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        216  .....                235

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        593  C.....                570

```

>[LC793011.1](#) Lecanicillium sp. CKSI15-F10-E2-i1 genes for 5.8S rRNA, ITS2 and 28S rRNA, partial and complete sequence

```

product length = 355
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        209  .....                228

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        563  C.....                540

```

>[MZ930389.1](#) Metarhizium anisopliae isolate UPASI_4 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        131  .....                150

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24
Template        497  C.....                474

```

>[OP900555.1](#) Fusarium solani isolate G9 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer  1    GCATCGATGAAGAACGCAGC   20
Template        212  .....                231

Reverse primer  1    NTTTTCCTCCGCTTATTGATATGC   24

```

Template 571 -..... 549

>[MW791926.1](#) *Fusarium petersiae* voucher culture Y224T internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 355

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 188 207

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 542 C..... 519

>[EU307898.1](#) *Metarhizium anisopliae* var. *anisopliae* strain LRC 182 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 366

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 530 549

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 895 C..... 872

>[OQ449708.1](#) *Fusarium proliferatum* isolate NBRIGKF-8 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 82 101

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 435 C..... 412

>[HF546329.2](#) *Geosmithia* sp. CCF3355 genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain CCF3355, isolate U416

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 221 240

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 585 C..... 562

>[OK560832.1](#) *Trichoderma atrobrunneum* strain TR09 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 274 293

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 634 -..... 612

>[MT557156.1](#) *Trichoderma brevicompactum* isolate RL177 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 357

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 199 218

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 555 C..... 532

>[MK406572.1](#) Uncultured *Fusarium* clone D768ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 233 252

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 586 C..... 563

>[HE649416.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT19

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 209 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 570 C..... 547

>[OM106601.1](#) *Fusarium* sp. isolate GLBRC404 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 362

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 196 215

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 557 C..... 534

>[PX350055.1](#) *Fusarium* sp. isolate LLC1482 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 351

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 220 239

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 570 C..... 547

>[GU480957.1](#) *Fusarium sacchari* isolate HMA-20 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S

ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 106 125

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 448 C..... 425

>[JX231150.1](#) Ilyonectria sp. LL-2013a culture-collection CPC:20693 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 570 C..... 547

>[OQ421989.1](#) Fusarium sp. breed MHUB11408 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 198 217

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 539 C..... 516

>[MW659186.1](#) Fusarium incarnatum strain F19 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 180 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 520 -..... 498

>[MG274309.1](#) Fusarium langsethiae isolate 0420 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 341

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 226 245

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 566 C..... 543

>[MK408022.1](#) Uncultured Fusarium clone D2253ITS small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       232 .....                251

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       585 C.....                562

```

>[AB378545.1](#) *Metacordyceps chlamydosporia* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, isolate: ID05-F0165

```

product length = 384
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       232 .....                251

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       615 C.....                592

```

>[ON492443.1](#) *Aciculosporium oplismeni* strain SYN217 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       245 .....                264

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       603 C.....                580

```

>[ON411513.1](#) *Fusarium proliferatum* isolate T5.6 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       167 .....                186

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       521 C.....                498

```

>[PQ517213.1](#) *Ilyonectria cyclaminicola* strain elc small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1   GCATCGATGAAGAACGCAGC   20
Template       193 .....                212

Reverse primer 1   NTTTTCTCCGCTTATTGATATGC   24
Template       546 C.....                523

```

>[KU845885.1](#) *Albifimbria verrucaria* strain 207C9 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1   GCATCGATGAAGAACGCAGC   20

```

```

Template      251 ..... 270

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      611 C..... 588

```

>[MT557412.1](#) *Fusarium* sp. isolate RL385 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 342
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      143 ..... 162

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      484 C..... 461

```

>[MW298933.1](#) *Hypocreales* sp. isolate 836467-30--2 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 362
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      175 ..... 194

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      536 C..... 513

```

>[NR_121429.1](#) *Hypomyces gabonensis* TFC 201156 ITS region; from TYPE material

```

product length = 378
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      210 ..... 229

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      587 C..... 564

```

>[OU720016.1](#) *Claviceps maximensis* genomic DNA sequence contains ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene, strain CCC1468

```

product length = 375
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      168 ..... 187

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      542 C..... 519

```

>[EF417482.1](#) *Trichoderma atroviride* small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 367
Forward primer 1   GCATCGATGAAGAACGCAGC  20
Template      267 ..... 286

Reverse primer 1   NTTTTCCTCCGCTTATTGATATGC  24
Template      633 C..... 610

```

>[LC681716.1](#) *Claviceps yanagawaensis* TNS:F-96440 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 381

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 246 265

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 626 C..... 603

>[OM515045.1](#) *Trichoderma azevedoi* strain CEN1304 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 363

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 349 368

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 711 C..... 688

>[HF546295.1](#) *Geosmithia* sp. U119A genomic DNA containing 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2 and 28S rRNA gene, strain U119A, isolate U218

product length = 365

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 223 242

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 587 C..... 564

>[FJ765022.1](#) *Purpureocillium lilacinum* isolate PAK-2 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 354

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 262 281

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 615 C..... 592

>[OW987479.1](#) *Purpureocillium lilacinum* genomic DNA sequence contains 18S rRNA gene, ITS1, 5.8S rRNA gene, ITS2, 28S rRNA gene

product length = 352

Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 264 283

Reverse primer 1 NTTTCCTCCGCTTATTGATATGC 24
 Template 615 C..... 592

>[MZ423618.1](#) *Nectriaceae* sp. isolate GW20_151 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 352
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 169 ..... 188

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 520 C..... 497

```

>[NR_121424.1](#) *Cladobotryum paravirescens* TFC 97-23 ITS region; from TYPE material

```

product length = 370
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 ..... 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 575 C..... 552

```

>[KP972551.1](#) *Calonectria* sp. CAK-1 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 354
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 281 ..... 300

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 634 C..... 611

```

>[PP897821.1](#) *Fusarium proliferatum* isolate ITS391 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 177 ..... 196

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 531 C..... 508

```

>[MN644580.1](#) *Trichoderma afroharzianum* voucher research collection Farrer lab 110 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 209 ..... 228

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 571 C..... 548

```

>[OR594352.1](#) *Sarocladium zeae* isolate HSRSZ6 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 327 ..... 346

```

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 685 C..... 662

>[JN689380.1](#) *Sarocladium oryzae* strain CBE-5 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2, partial sequence

product length = 361
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 202 221

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 562 T..... 539

>[PX350743.1](#) *Fusarium* sp. isolate LLC2981 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 217 236

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 558 C..... 535

>[KC806295.1](#) *Clonostachys* sp. LMA-2013 strain CML 1910 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 361
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 176 195

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 536 C..... 513

>[JN195741.1](#) *Beauveria bassiana* isolate F. Carrap. 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 351
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 310 329

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 660 C..... 637

>[AB360364.1](#) *Lecanicillium psalliotae* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence, strain: KYK00175

product length = 354
 Forward primer 1 GCATCGATGAAGAACGCAGC 20
 Template 641 660

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
 Template 994 C..... 971

>[ON207931.1](#) *Fusarium poae* strain CN062F1 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 341
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       310    ..... 329

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       650    C..... 627
```

>[MW299458.1](#) *Hypocreales* sp. isolate 928717-31 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 361
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       140    ..... 159

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       500    C..... 477
```

>[KY607544.1](#) *Ijuhya peristomialis* strain CBS 569.76 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene, complete sequence; and internal transcribed spacer 2', partial sequence

```
product length = 339
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       368    ..... 387

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       706    C..... 683
```

>[KC330218.1](#) *Trichoderma harzianum* strain CKP01 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```
product length = 363
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       339    ..... 358

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       701    C..... 678
```

>[MG274296.1](#) *Fusarium tricinctum* isolate 07015 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```
product length = 355
Forward primer 1      GCATCGATGAAGAACGCAGC  20
Template       227    ..... 246

Reverse primer 1      NTTTCCTCCGCTTATTGATATGC  24
Template       581    C..... 558
```

>[HE649446.1](#) *Hypocrea lixii* genomic DNA containing ITS1, 5.8S ribosomal RNA gene and ITS2, and 28S ribosomal RNA gene, isolate TUT86

```

product length = 363
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 206 ..... 225

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 568 C..... 545

```

>[MZ423663.1](#) Nectriaceae sp. isolate GW20_296 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 361
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 183 ..... 202

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 543 C..... 520

```

>[PV997965.1](#) Fusarium solani isolate LJMh-2-2-1-1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 360
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 180 ..... 199

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 538 -..... 516

```

>[MK761187.1](#) Fusarium oxysporum internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 160 ..... 179

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 514 C..... 491

```

>[ON411507.1](#) Fusarium proliferatum isolate T4.1 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 355
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 123 ..... 142

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 477 C..... 454

```

>[KC806259.1](#) Clonostachys pseudocholeuca strain CML 824 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 357

```

```

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 175 ..... 194

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 531 C..... 508

```

>[LC681689.1](#) *Claviceps purpurea* TNS:F-96524 genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

```

product length = 369
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 210 ..... 229

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 578 C..... 555

```

>[ON006797.1](#) *Sarocladium zeae* strain CN030A7 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

```

product length = 359
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 330 ..... 349

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 688 C..... 665

```

>[JN235158.1](#) *Fusarium keratoplasticum* strain FRC S-2412 5.8S ribosomal RNA gene, complete sequence

```

product length = 362
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 182 ..... 201

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 543 C..... 520

```

>[KM242298.1](#) *Eucasphaeria* sp. INBio_45030 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 378
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 32 ..... 51

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 409 C..... 386

```

>[JN093261.1](#) *Verticillium* sp. NRRL 46126 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

```

product length = 350
Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 151 ..... 170

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24

```

Template 500 C..... 477

>[KX681579.1](#) *Fusarium subglutinans* isolate M02S_A1_92 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 149 168

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 491 C..... 468

>[FJ765025.1](#) *Paecilomyces marquandii* isolate CNSH 18S ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and 28S ribosomal RNA gene, partial sequence

product length = 380

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 211 230

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 590 C..... 567

>[OQ248230.1](#) *Fusarium incarnatum* strain CoLD Pav_14 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 294 313

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 635 C..... 612

>[MW131978.1](#) *Fusarium equiseti* strain BNHai small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 223 242

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 564 C..... 541

>[OP897211.1](#) *Fusarium equiseti* isolate NP_OH1827_2 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 342

Forward primer 1 GCATCGATGAAGAACGCAGC 20
Template 307 326

Reverse primer 1 NTTTTCTCCGCTTATTGATATGC 24
Template 648 C..... 625

>[AB027372.1](#) *Cordyceps ramosopulvinata* genes for 18S rRNA, ITS1, 5.8S rRNA, ITS2, 28S rRNA, partial and complete sequence

product length = 413

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	448		467

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	860	C.....	837

>[PV108355.1](#) *Fusarium verticillioides* strain CIRM-BRFM 1385 small subunit ribosomal RNA gene, partial sequence; internal transcribed spacer 1, 5.8S ribosomal RNA gene, and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 343

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	314		333

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	656	C.....	633

>[OM106581.1](#) *Fusarium* sp. isolate GLBRC384 internal transcribed spacer 1, partial sequence; 5.8S ribosomal RNA gene and internal transcribed spacer 2, complete sequence; and large subunit ribosomal RNA gene, partial sequence

product length = 361

Forward primer	1	GCATCGATGAAGAACGCAGC	20
Template	183		202

Reverse primer	1	NTTTCCTCCGCTTATTGATATGC	24
Template	543	C.....	520

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